

Clustering

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Outline

- ▶ What is Clustering?
- ▶ Types of Data to Cluster
- ▶ Distance/ Similarity Measures
- ▶ Clustering Methods
 - ▶ K-means and K-means++
 - ▶ K-medoids
 - ▶ Hierarchical clustering method
 - ▶ DBSCAN
- ▶ Cluster Validity Measures

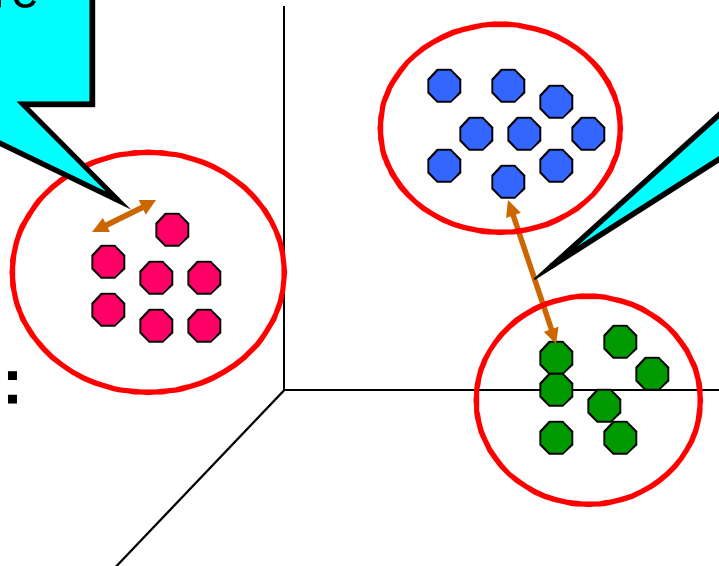
What Is Clustering?

- ▶ Group data into clusters so that the points in one group are **similar** to each other and are as **different** as possible from the points in other groups

Intra-cluster distances are minimized

Inter-cluster distances are maximized

**Unsupervised learning:
no predefined classes**



Examples of Clustering Applications (1)

- ▶ **Customer Segmentation (Marketing):**
 - ▶ **Grouping customers** based on purchasing behavior, preferences, and demographics to tailor marketing strategies and product offerings.
- ▶ **Document Clustering (Text Mining):**
 - ▶ **Grouping similar documents** together, which is useful for organizing large document corpora, such as news articles, research papers, or social media posts.
- ▶ **Image Segmentation (Computer Vision):**
 - ▶ **Dividing an image into segments** for easier analysis, such as object recognition or medical image analysis.
 - ▶ E.g., Soil scientists filter trees from background

Example of Image Segmentation

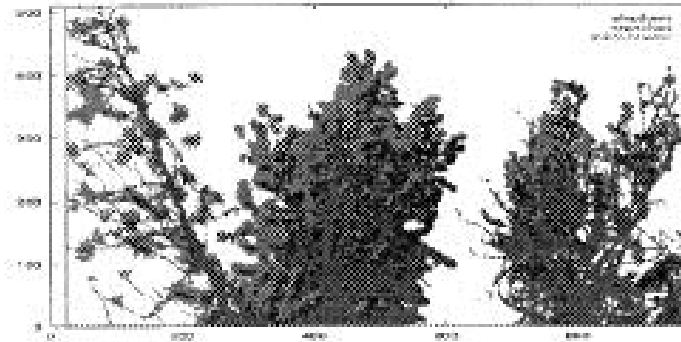
- Filtering real images
 - Images of trees taken in near-infrared band (NIR) and visible wavelength (VIS)
 - 512x1024 pixels and each of them contains a pair of brightness values (NIR,VIS)



The images taken in NIR and VIS

1st step: 284 seconds

2nd step: 71 seconds



The sunlit leaves, branches and shadows

Examples of Clustering Applications (2)

▶ **Bioinformatics & Healthcare**

- ▶ Gene expression analysis (**grouping similar genes** to predict possible functions of genes with unknown function)
- ▶ Disease outbreak detection (**clustering similar symptoms or patients**)
- ▶ Drug discovery (**clustering similar chemical compounds**)

▶ **Social Network Analysis:**

- ▶ **Grouping users based on their interactions**, such as identifying communities or clusters of individuals with similar interests in social networks.

▶ **Recommendation Systems:**

- ▶ **Grouping users or items based on preferences** to make personalized recommendations, for instance in movie recommendations or e-commerce platforms.

Examples of Clustering Applications (3)

▶ **City-planning:**

- ▶ Identifying groups of houses according to their house type, value, and geographical location

▶ **WWW**

- ▶ Cluster web documents
- ▶ Cluster web log data to discover groups of users

▶ **Anomaly Detection:**

- ▶ Identifying outliers or abnormal behavior in data, such as fraud detection in financial transactions or network intrusion detection.

Types of Data to Be Clustered (1)

- ▶ Tabular data (Data matrix)

- ▶ Each row is an data object (called example)
 - ▶ E.g., a customer, a document, an image, a data point in Euclidean space, etc.
- ▶ Each column represents an attribute or feature

Customer ID	Age	Income	Gender	Membersh ip Type
101	25	50000	Male	Gold
102	32	62000	Female	Silver
103	45	75000	Male	Platinum
104	29	48000	Female	Gold
105	36	70000	Male	Silver

- ▶ No class labels. Learning is *unsupervised*.

Types of Data to Be Clustered (2)

▶ Sets

- ▶ Documents, each represented by e.g., a set of n-grams,
- ▶ Movies:

Movie ID	Genre Set
1	{Action, Adventure, Sci-Fi}
2	{Drama, Romance}
3	{Action, Thriller, Crime}
4	{Comedy, Family}
5	{Adventure, Fantasy, Action}
6	{Drama, Crime, Mystery}

▶ Sequences

- ▶ genes, click streams on a web site, time-series, etc.

▶ Graphs

- ▶ Protein interaction networks, chemical compounds, etc.

Attribute Types of Tabular Data

▶ Example of data matrix

Customer ID	Age	Income	Gender	Membership Type
101	25	50000	Male	Gold
102	32	62000	Female	Silver
103	45	75000	Male	Platinum
104	29	48000	Female	Gold
105	36	70000	Male	Silver

▶ Attributes can be

- ▶ Numeric (real values, integers)
- ▶ Categorical (Symbolic)
 - ▶ Nominal (values without meaningful order)
 - ▶ Color, gender, product types
 - ▶ Ordinal (values with a meaningful order)
 - ▶ Temperature (low, medium, high)

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- ▶ Distance/ Similarity Measures
- ▶ Clustering Methods
 - ▶ K-means and K-means++
 - ▶ K-medoids
 - ▶ Hierarchical clustering method
- ▶ Disk-based large-scale clustering algorithms
 - ▶ BFR
 - ▶ CURE
- ▶ Cluster Validity Measures

Commonly Used Distance Measures

- ▶ Minkowski Distance (General Form)
 - ▶ For numeric data
- ▶ Cosine Distance
 - ▶ For numeric data
- ▶ Jaccard Distance
 - ▶ For sets
- ▶ Hamming Distance
 - ▶ For strings of the same lengths

Distance Metric

- ▶ A distance measure is a ***distance metric*** if it satisfies the following 4 conditions:
 - ▶ **Non-negativity:**
$$d(x,y) \geq 0$$
 - ▶ **Identity of Indiscernibles:**
$$d(x,y) = 0 \text{ if and only if } x=y$$
 - ▶ **Symmetry:**
$$d(x,y) = d(y,x)$$
 - ▶ **Triangle Inequality:**
$$d(x,z) \leq d(x,y) + d(y,z)$$
- ▶ For robust clustering, always prefer metrics

Distance (or Similarity) Measures

- ▶ For data containing only *numeric attributes*, a popular measure of distance between two data objects, i and j , is *Minkowski distance*:

$$d(i, j) = \sqrt[q]{(|x_{i1} - x_{j1}|^q + |x_{i2} - x_{j2}|^q + \dots + |x_{ip} - x_{jp}|^q)}$$

where $i = (x_{i1}, x_{i2}, \dots, x_{ip})$ and $j = (x_{j1}, x_{j2}, \dots, x_{jp})$ are two p -dimensional data objects, and q is a positive integer.

- ▶ If $q = 1$, d is *Manhattan distance*

$$d(i, j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \dots + |x_{ip} - x_{jp}|$$

Distance (or Similarity) Measures

- ▶ If $q = 2$, d is Euclidean distance:

$$d(i, j) = \sqrt{(x_{i_1} - x_{j_1})^2 + (x_{i_2} - x_{j_2})^2 + \dots + (x_{i_p} - x_{j_p})^2}$$

- ▶ **Minkowski distance is a distance metric**

- ▶ $d(i, j) \geq 0$ (nonnegative)
- ▶ $d(i, i) = 0$ (distance to itself is 0, and iff the two objects are same)
- ▶ $d(i, j) = d(j, i)$ (symmetric)
- ▶ $d(i, j) \leq d(i, k) + d(k, j)$ (Triangular inequality)

- ▶ Also, one can use weighted distance (or other dissimilarity measures)

$$d(i, j) = \sqrt[q]{w_1 |x_{i_1} - x_{j_1}|^q + w_2 |x_{i_2} - x_{j_2}|^q + \dots + w_p |x_{i_p} - x_{j_p}|^q} \quad (q > 0)$$

Distance (or Similarity) Measures

- ▶ Another similarity measure for data with only numeric attributes is cosine similarity:

$$\text{Similarity}(\mathbf{x}, \mathbf{y}) = \cos(\theta) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$$

where θ is the angle between two examples $\mathbf{x}$ and $\mathbf{y}$.

- ▶ **Cosine Distance** = 1- Cosine_Similarity
- ▶ Cosine distance is not a distance metric. It does not satisfy
 - ▶ Identity of indiscernibles
 - ▶ Triangle inequality

Commonly Used Distance Measures

▶ Jaccard Distance

- ▶ Jaccard similarity measures the similarity between two sets: S_1 and S_2 :

$$Sim(S_1, S_2) = |S_1 \cap S_2| / |S_1 \cup S_2|$$

- ▶ Jaccard distance = $1 - Sim(S_1, S_2)$
- ▶ It is a distance metric

▶ Hamming Distance

- ▶ Measures how different two strings of equal length are.
- ▶ Counts the number of positions at which the corresponding symbols (usually bits or characters) are different.

- ▶ Example:

- ▶ String A: 1011101

- ▶ String B: 1001001

Distance(A, B) = 2

- ▶ It is a distance metric

Distance Measures

- ▶ We focus on tabular data with Minkowski distance measure
- ▶ It is good for numerical data.
- ▶ What if data contains only *nominal attributes*?
- ▶ Method 1:
 - ▶ Convert a nominal attribute to a numerical attribute, for example:
 - ▶ yes $\Rightarrow$ 1 and no $\Rightarrow$ 0
 - ▶ blue $\Rightarrow$ 1, yellow $\Rightarrow$ 2, red $\Rightarrow$ 3, etc.
 - ▶ Problem?

Distance Measures

▶ Method 2:

▶ One-hot encoding for each attribute

- ▶ For example, if the domain of attribute A is {a1, a2, a3}, we convert it into 3 attributes to represent A.
 - ▶ If $A=a_1$, the values of the three attribute are 1, 0, 0.
 - ▶ If $A=a_2$, the values of the three attribute are 0, 1, 0.
 - ▶ If $A=a_3$, the values of the three attribute are 0, 0, 1.

Distance (or Similarity) Measures

► Method 3:

- Distance between two instances $\mathbf{x}$ and $\mathbf{y}$:

$$\text{Distance}(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^m \text{dist}(x_i, y_i)$$

where $\mathbf{x}=(x_1, x_2, \dots, x_m)$ and $\mathbf{y}=(y_1, y_2, \dots, y_m)$

$$\text{dist}(x_i, y_i) = \begin{cases} 0 & \text{if } x_i \text{ and } y_i \text{ are nominal and } x_i = y_i \\ 1 & \text{if } x_i \text{ and } y_i \text{ are nominal and } x_i \neq y_i \end{cases}$$

Distance Measures

- ▶ What if data contain *both numeric and nominal attributes*?
- ▶ Method 1:
 - ▶ One-hot encoding for each nominal attribute

- ▶ Method 2:

$$\text{Distance}(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^m \text{dist}(x_i, y_i)$$

where $\mathbf{x}=(x_1, x_2, \dots, x_m)$ and $\mathbf{y}=(y_1, y_2, \dots, y_m)$

$$\text{dist}(x_i, y_i) = \begin{cases} 0 & \text{if } x_i \text{ and } y_i \text{ are nominal and } x_i = y_i \\ 1 & \text{if } x_i \text{ and } y_i \text{ are nominal and } x_i \neq y_i \\ | \text{norm}(x_i) - \text{norm}(y_i) | & \text{if } x_i \text{ and } y_i \text{ are continuous} \end{cases}$$

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Major Clustering Approaches

- ▶ Partitioning algorithms: Construct various partitions and then evaluate them by some criterion
- ▶ Hierarchical algorithms: Create a hierarchical decomposition of the set of data (or objects) using some criterion
- ▶ Density-based: based on connectivity and density functions
- ▶ Model-based: A model is hypothesized for each of the clusters and the idea is to find the best fit of that model to each other

Partitioning Algorithms: Basic Concept

- ▶ Partition n objects into k clusters
 - ▶ Optimize the chosen partitioning criterion
- ▶ Global optimal: exhaustively enumerate all partitions
 - ▶ Infeasible in practice
- ▶ Heuristic methods: *k-means* and *k-medoids* algorithms
 - ▶ *k-means* (MacQueen'67): Each cluster is represented by the mean values of the objects in the cluster (as the cluster center, or centroid)
 - ▶ *k-medoids* or PAM (Partition around medoids) (Kaufman & Rousseeuw'87): Each cluster is represented by one of the objects in the cluster

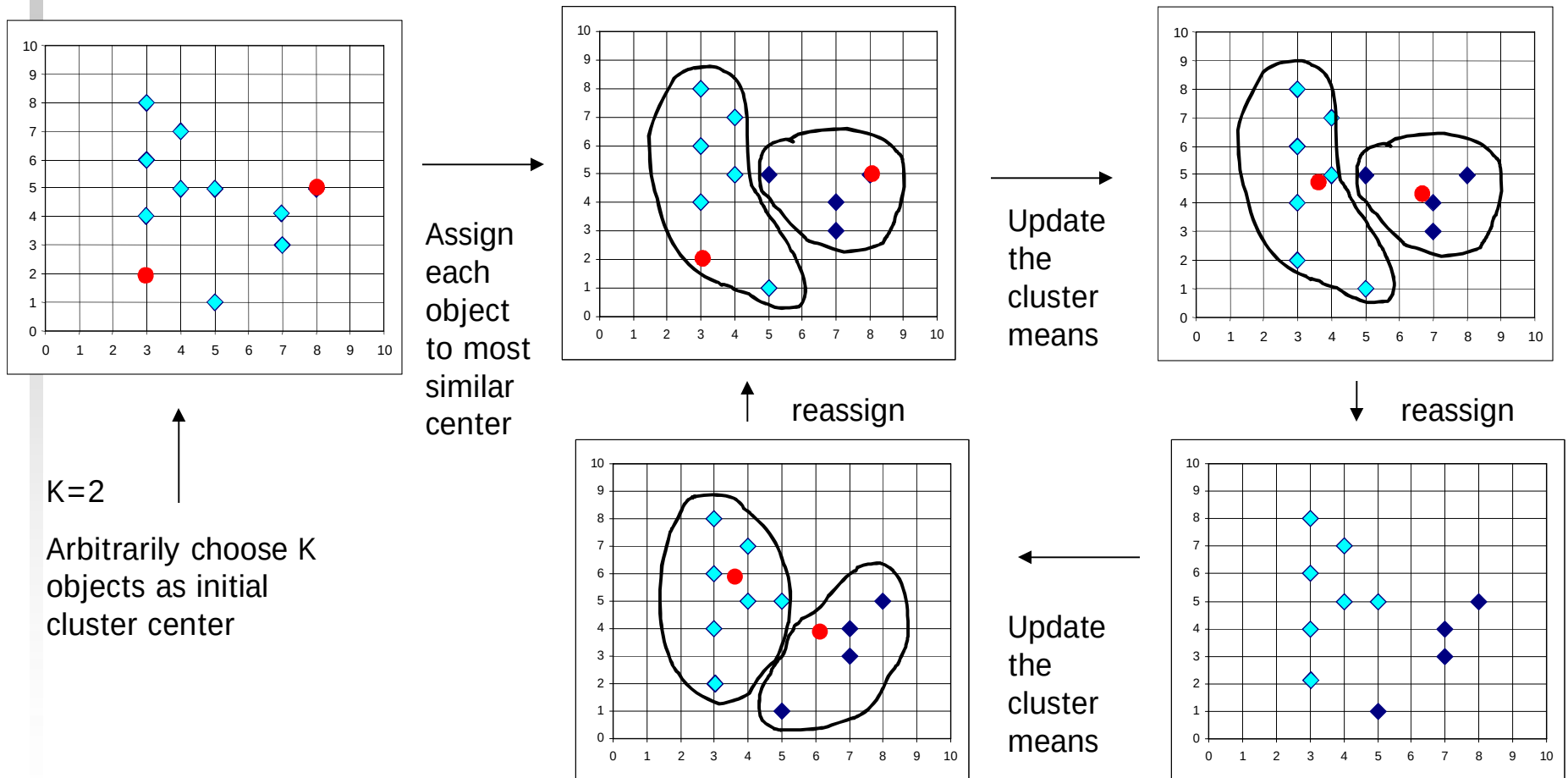
K-means

Group objects into k clusters:

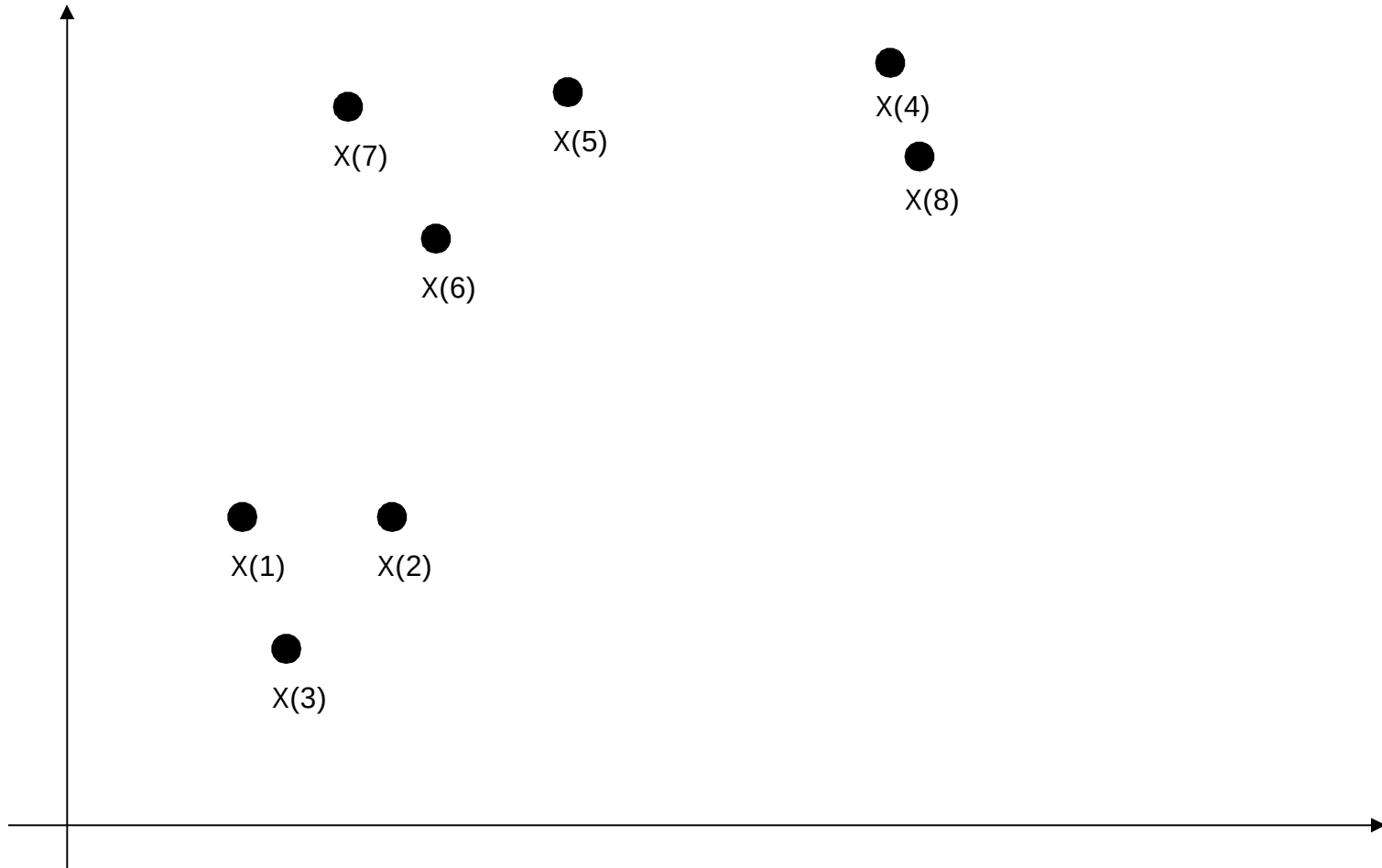
- ▶ Arbitrarily choose k objects as the initial cluster centers
- ▶ Until no change, do
 - ▶ (Re)assign each object to the cluster to which the object is the most similar, based on the mean value of the objects in the cluster
 - ▶ Update the cluster means, i.e., calculate the mean value of the objects for each cluster

The *K-Means* Clustering Method

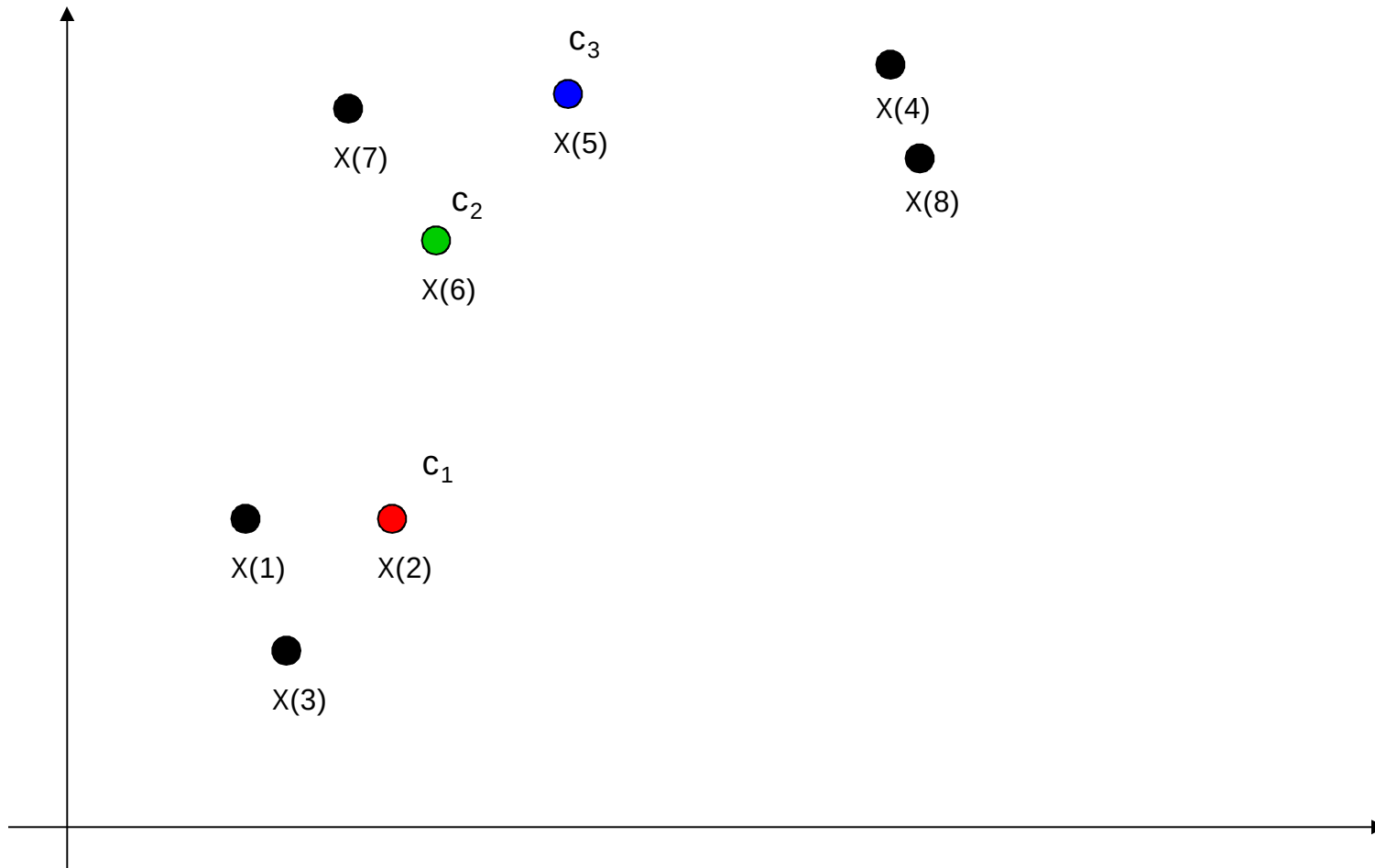
► Example



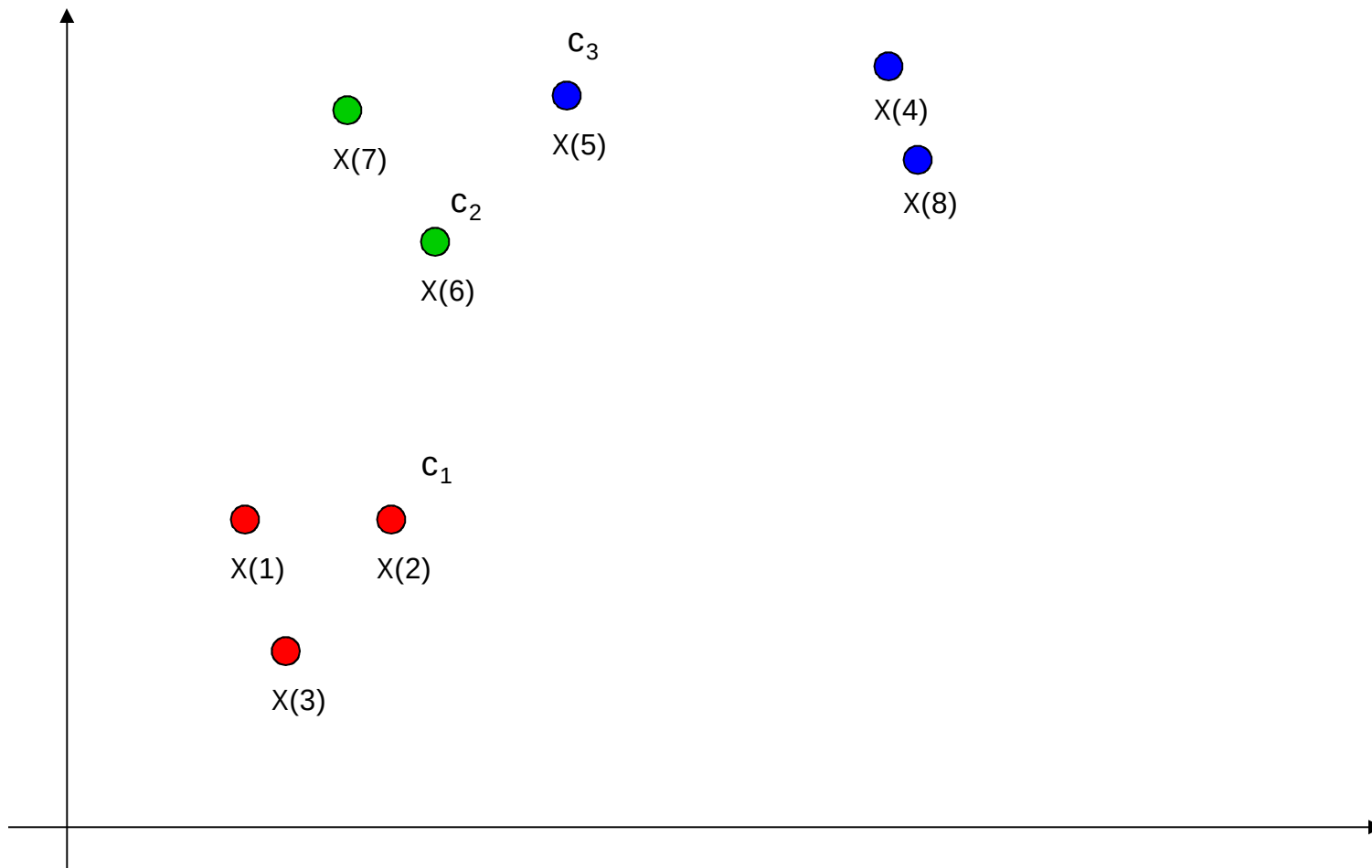
K-means example



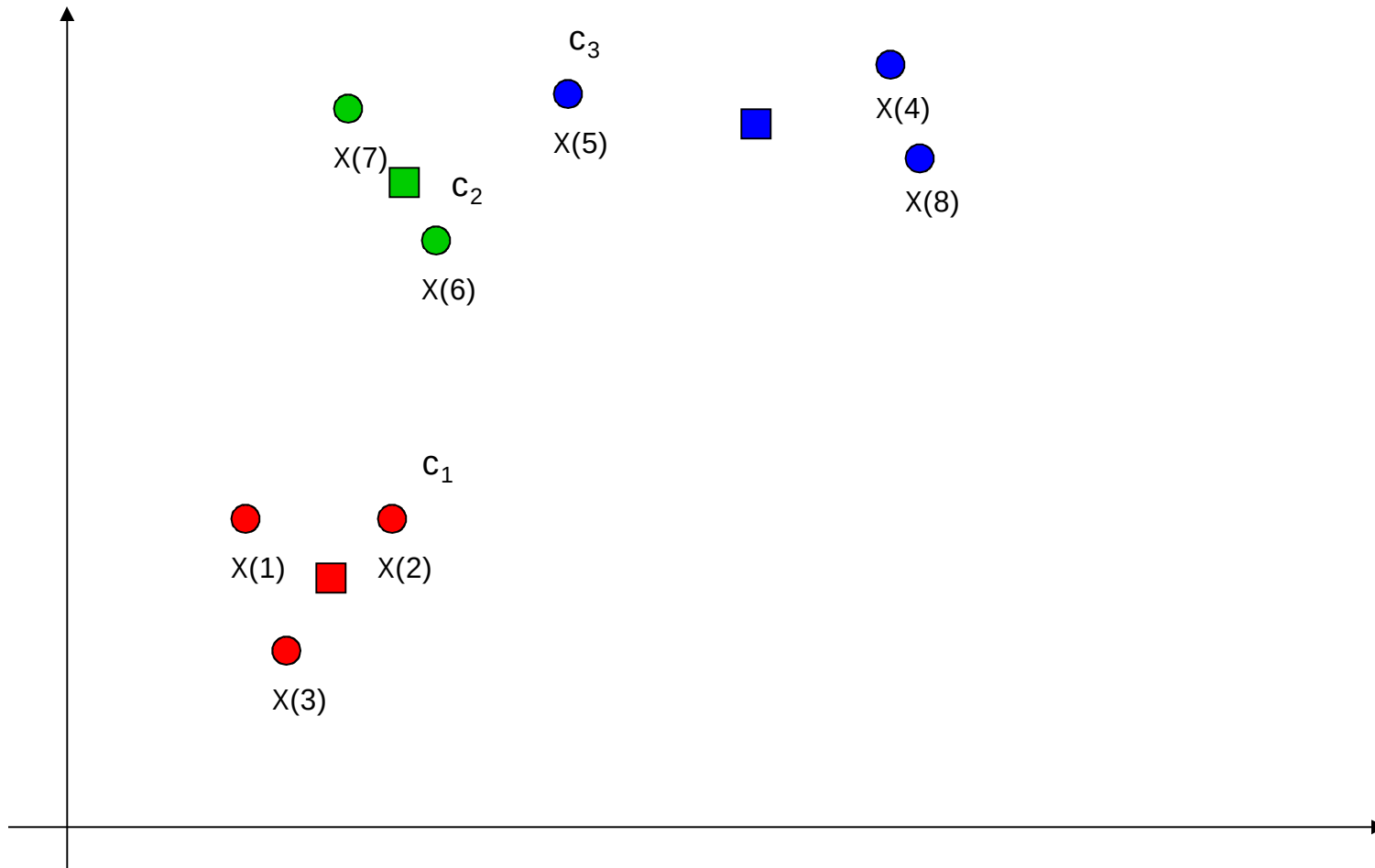
K-means example



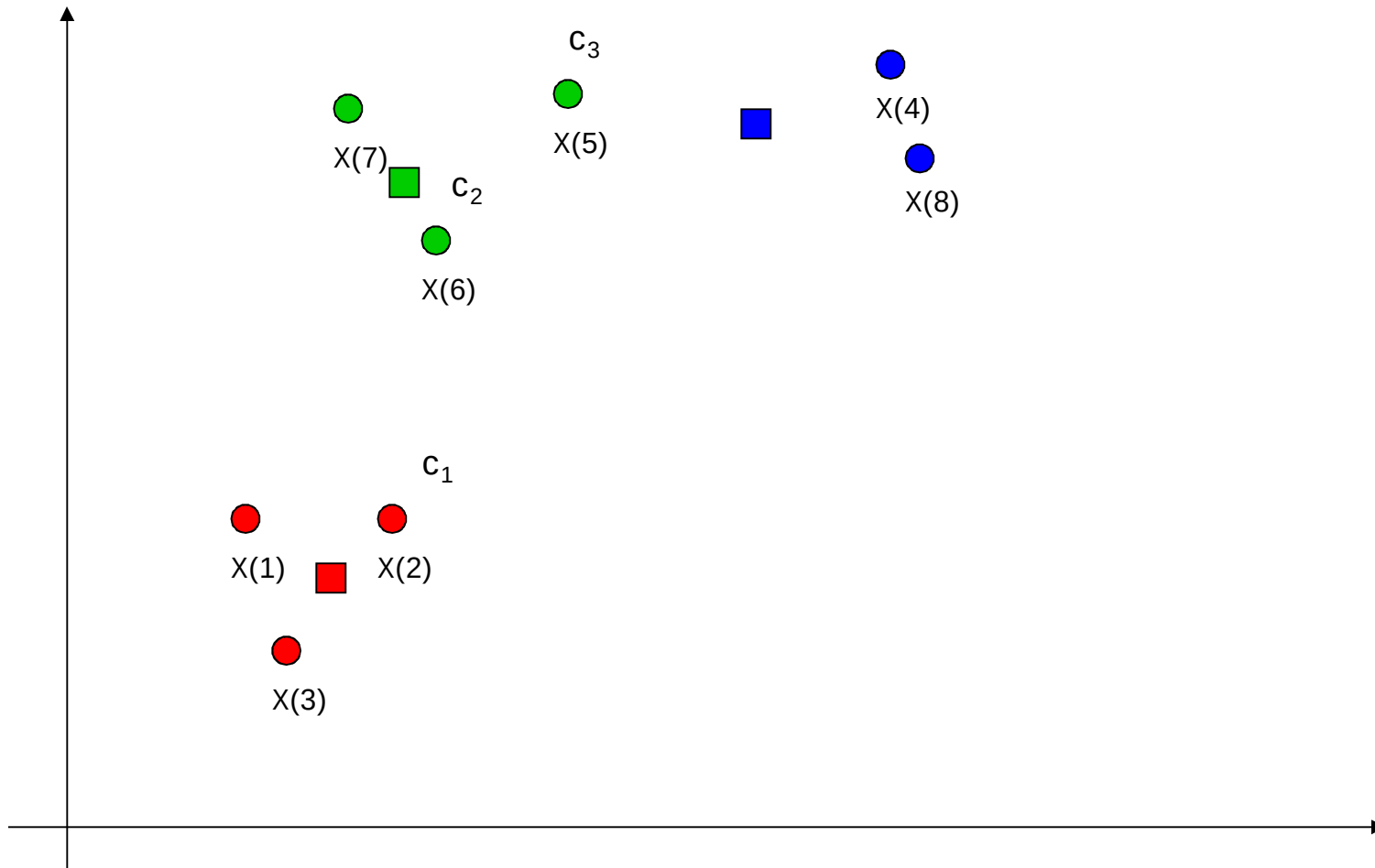
K-means example



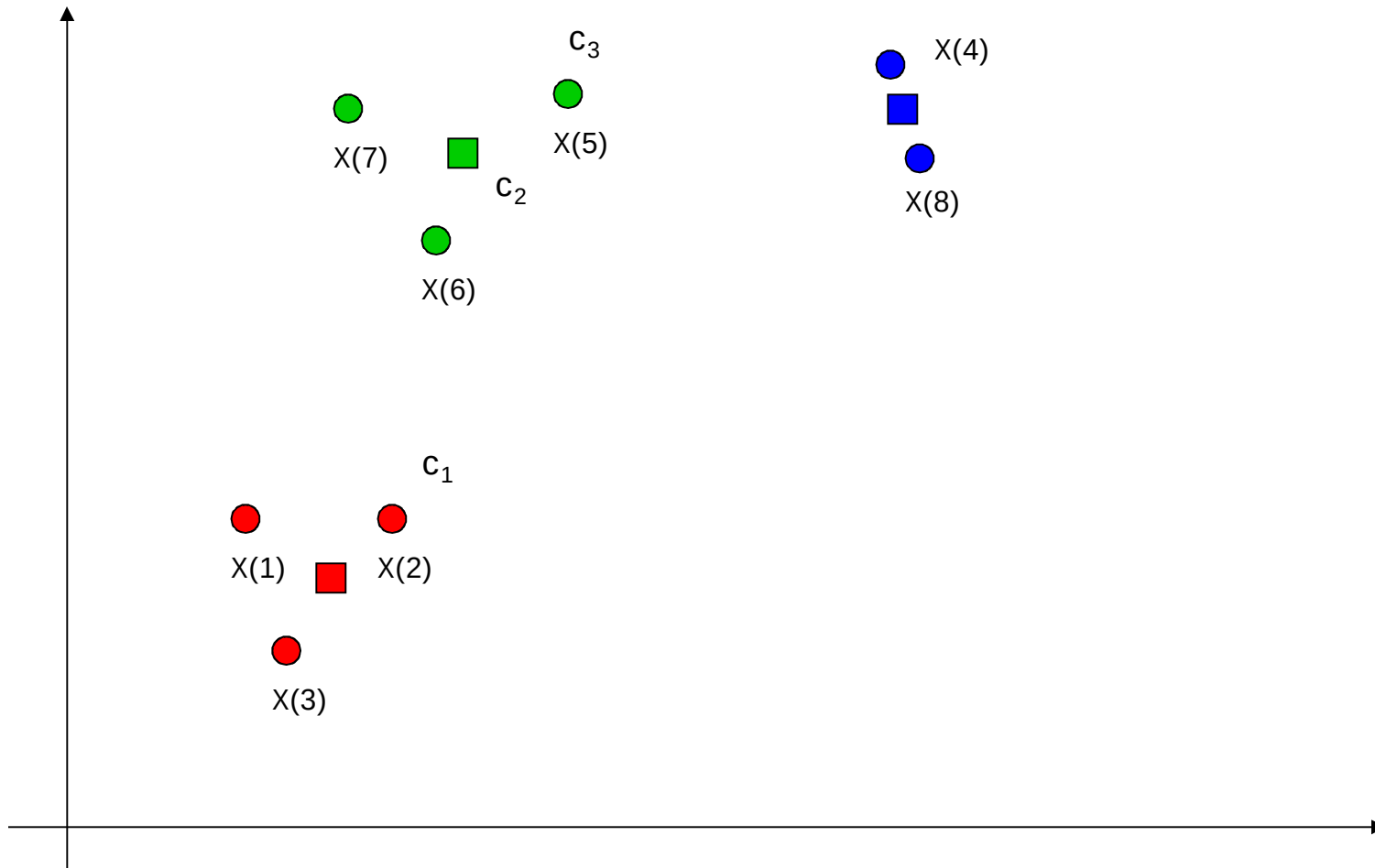
K-means example



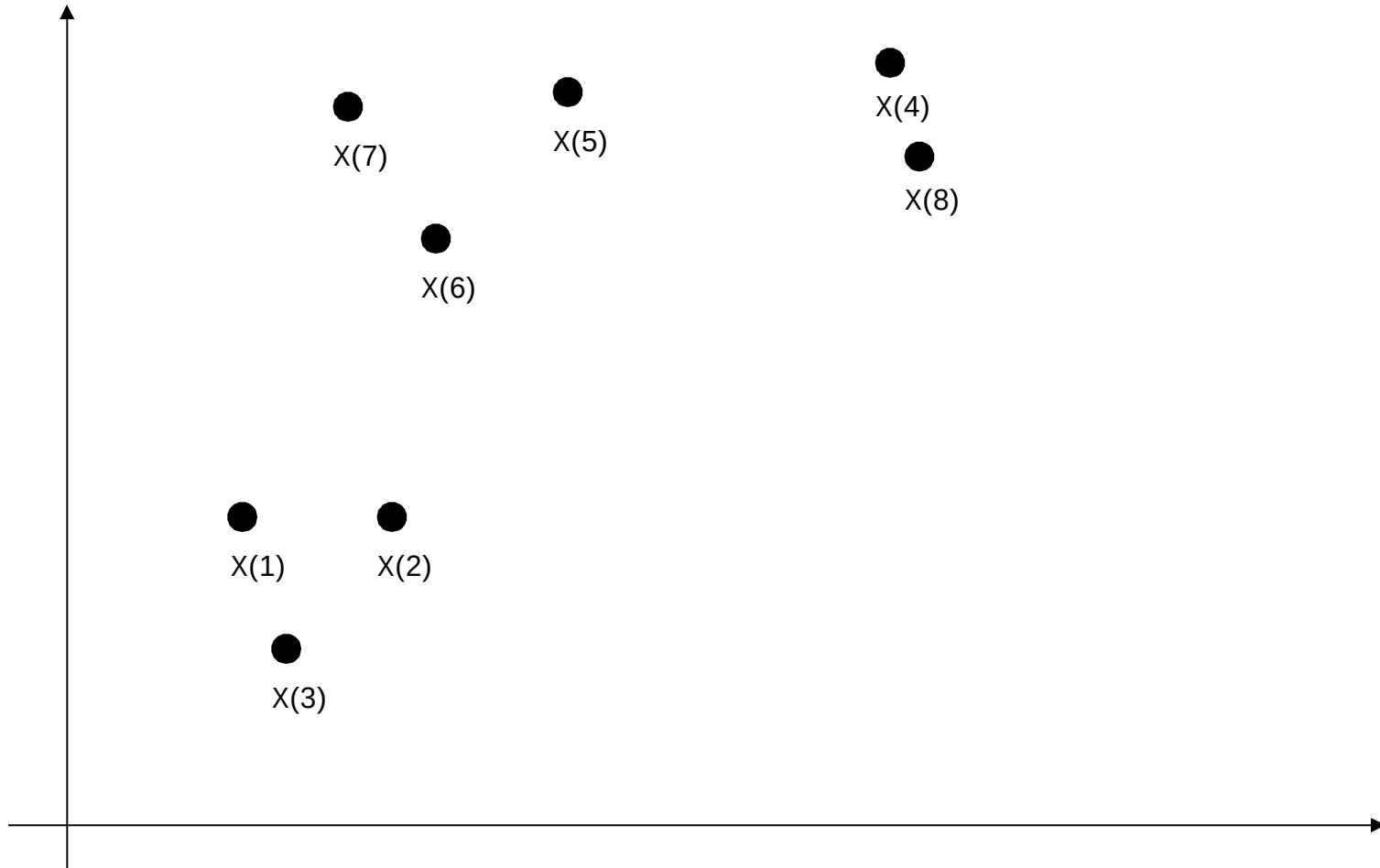
K-means example



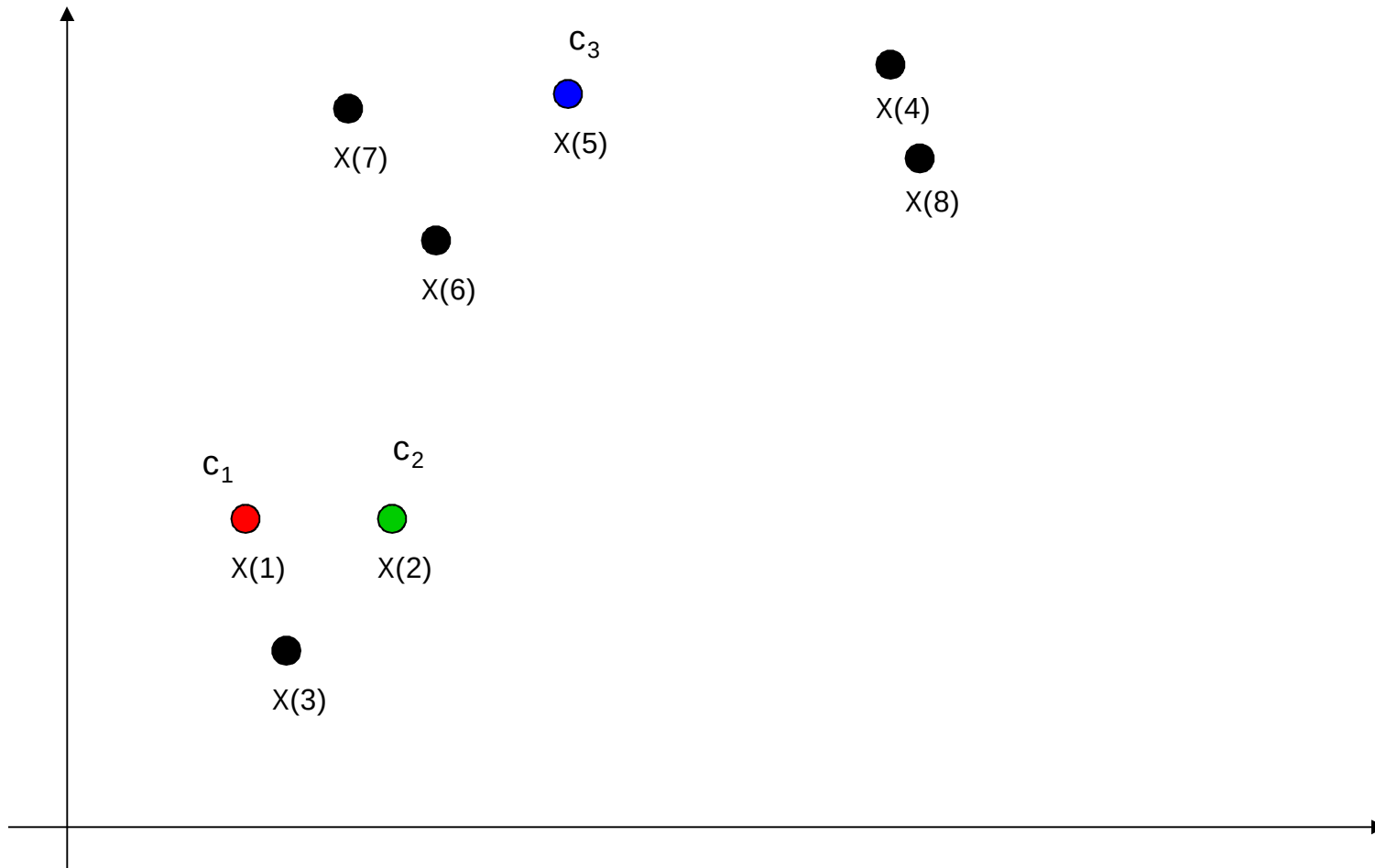
K-means example



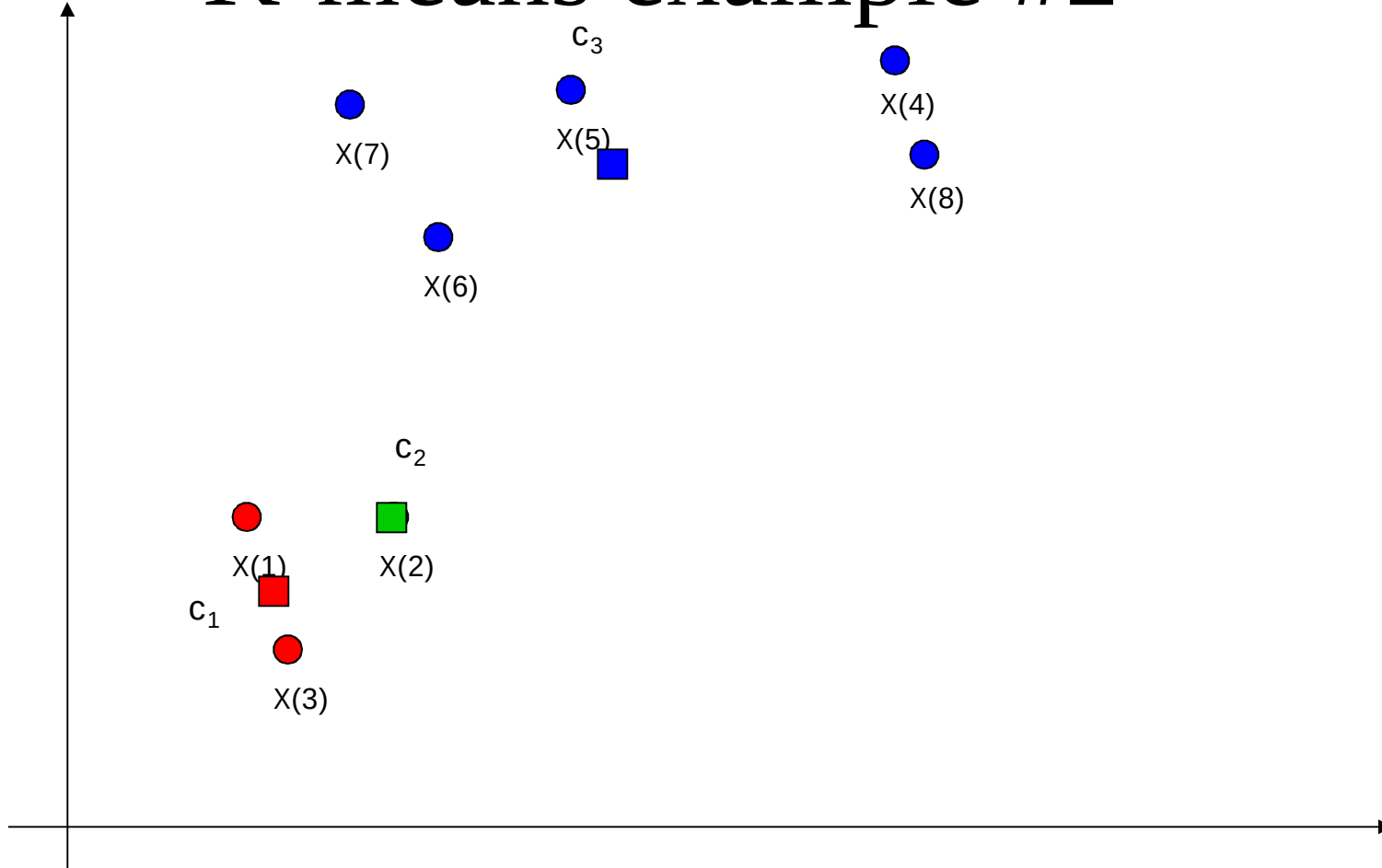
K-means example #2



K-means example #2



K-means example #2



How to Determine k

Several techniques. One of them is:

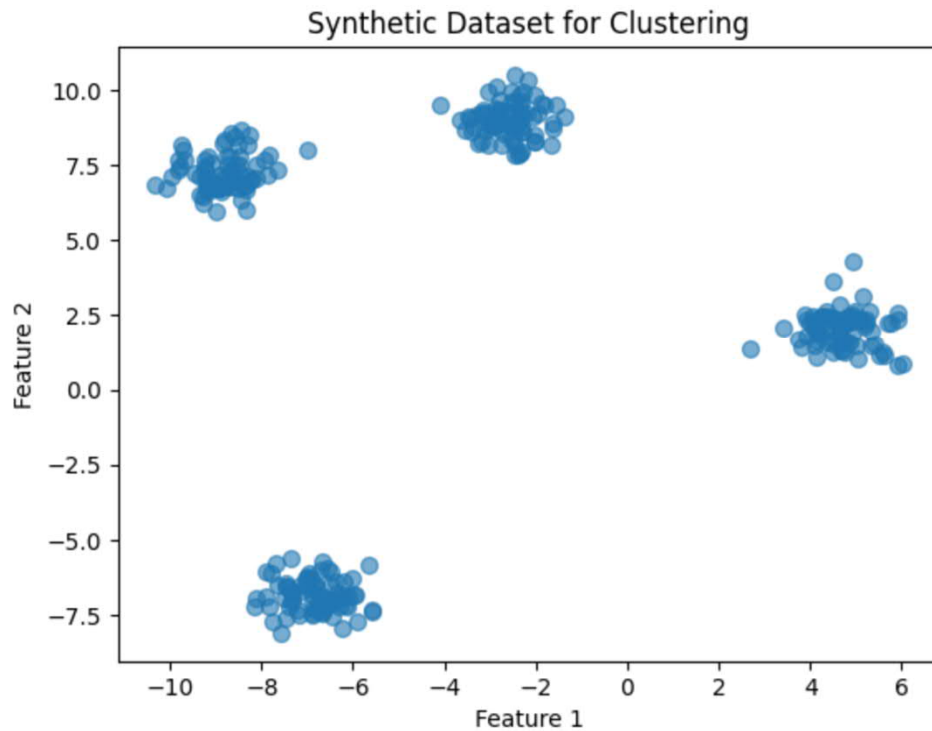
► Elbow Method

- Compute the *sum of squared distances (SSD)* between points and their assigned cluster centroids:

$$SSD = \sum_{i=1}^k \sum_{x \in C_i} dist(x, v_i)^2$$

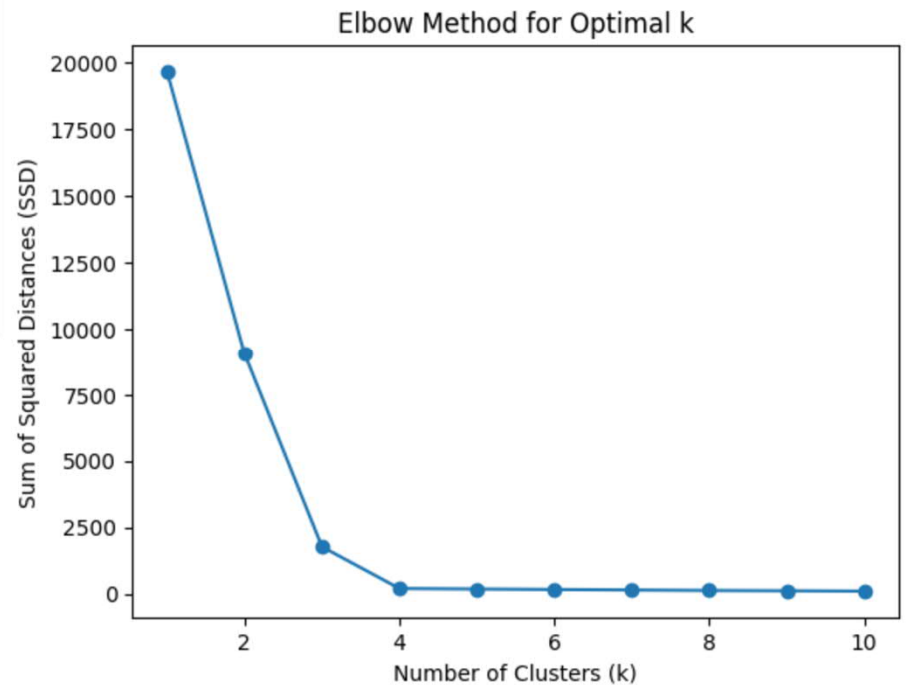
- v_i is the center of the cluster C_i
- k is the number of clusters
- Plot SSD against k
 - Look for an "elbow point" where the rate of decrease sharply slows, indicating diminishing benefits from adding more clusters.

Illustration of Elbow Method



The elbow is around **k=4**, matching the actual number of clusters in the synthetic dataset.

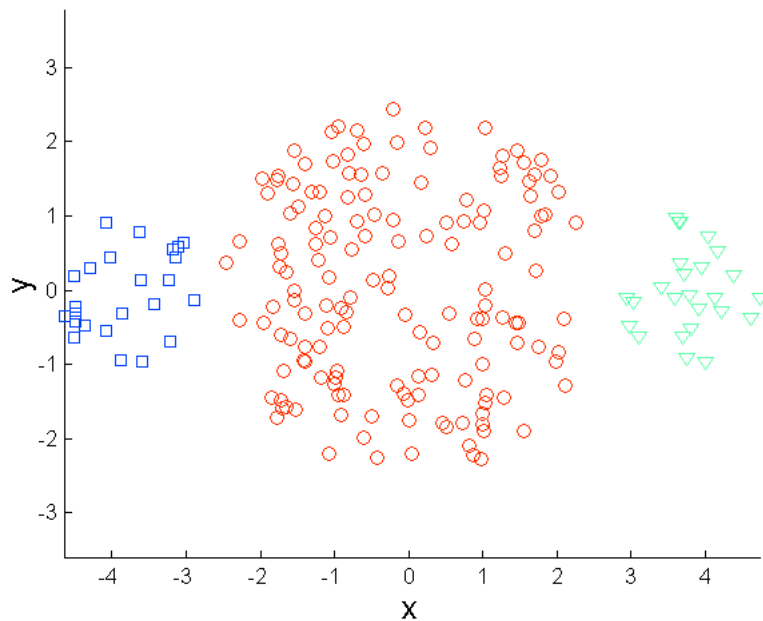
$$SSD = \sum_{i=1}^k \sum_{x \in C_i} dist(x, v_i)^2$$



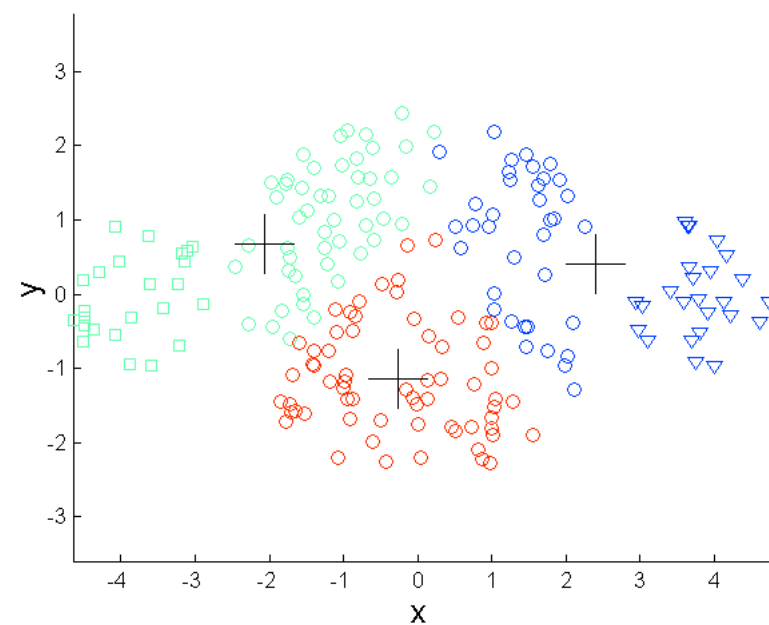
Comments on the *K-Means* Method

- ▶ Strength: *Relatively efficient*: $O(tkn)$, where n is # objects, k is # clusters, and t is # iterations. Normally, $k, t \ll n$.
 - ▶ Comparing: PAM: $O(k(n-k)^2)$, CLARA: $O(ks^2 + k(n-k))$
- ▶ Comment: Often terminates at a *local optimum*.
- ▶ Weaknesses
 - ▶ Sensitive to the initial clusters
 - ▶ Applicable only when *mean* is defined, then what about categorical data?
 - ▶ Need to specify k , the *number* of clusters, in advance
 - ▶ Very sensitive to noise and *outliers*
 - ▶ May have a problem when clusters have different sizes and are close to each other.  example
 - ▶ Not suitable to discover clusters with *non-convex shapes*  example

A Limitation of K-means: Differing Sizes

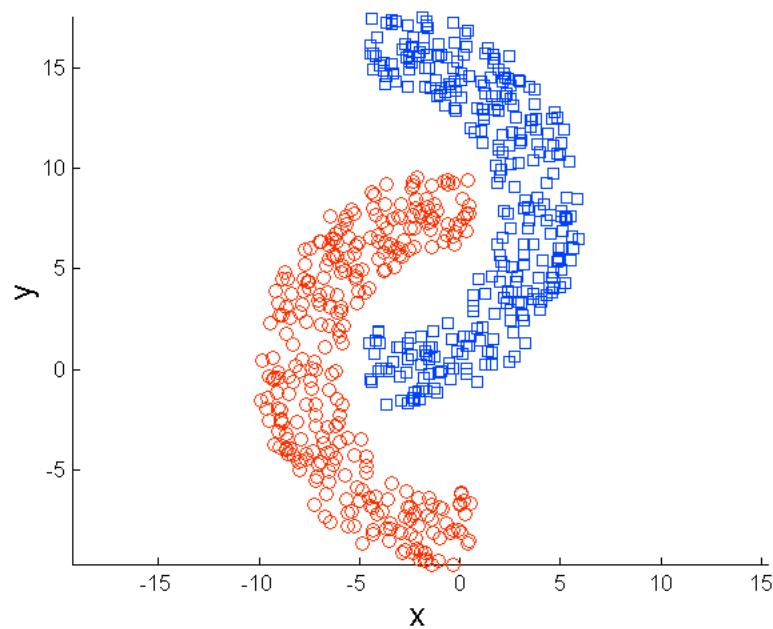


Original Points

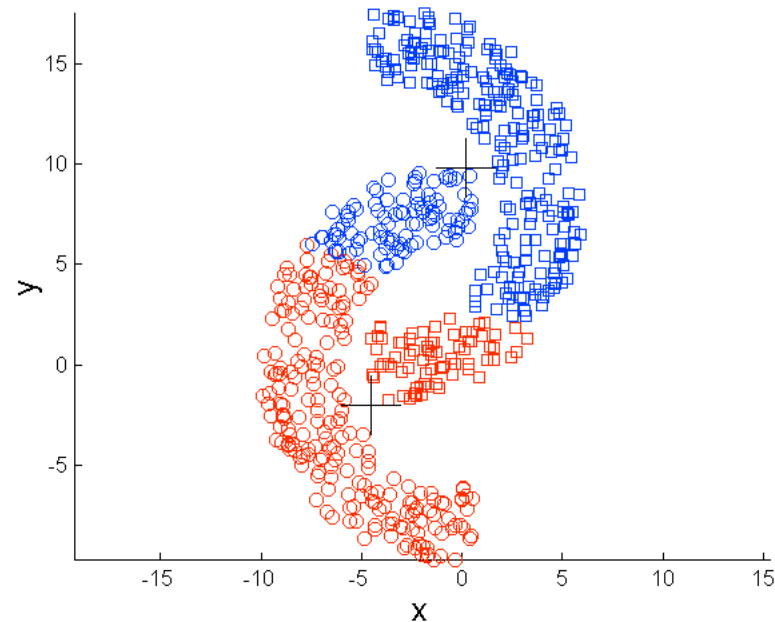


K-means (3 Clusters)

A Limitation of K-means: Non-globular (Non-Convex) Shapes



Original Points



K-means (2 Clusters)

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Better Initialization: k-Means++

- ▶ The k-means algorithm is sensitive to initial clusters
 - ▶ Not stable
 - ▶ Bad initialization leads to **bad-quality clusters** and typically **more iterations to converge**.
- ▶ K-Means++:
 - ▶ Uses a smarter initialization strategy
 - ▶ More stable
 - ▶ Improve clustering quality
 - ▶ Faster

k-Means++

1. Pick the first centroid randomly.
2. Select subsequent centroids probabilistically, favoring points farther from existing centroids.
3. Run k-means after initializing the centroids.

Selecting Subsequent Centroids Probabilistically (Step 2)

- ▶ Compute distance $D(x_i)$ for each data point

- ▶ For each point x_i in the data set X , compute:

$$D(x_i)^2 = \min_{c \in C} \|x_i - c\|^2$$

- C is the set of already chosen centroids
 - $\|x_i - c\|^2$ is the squared Euclidean distance between x_i and c

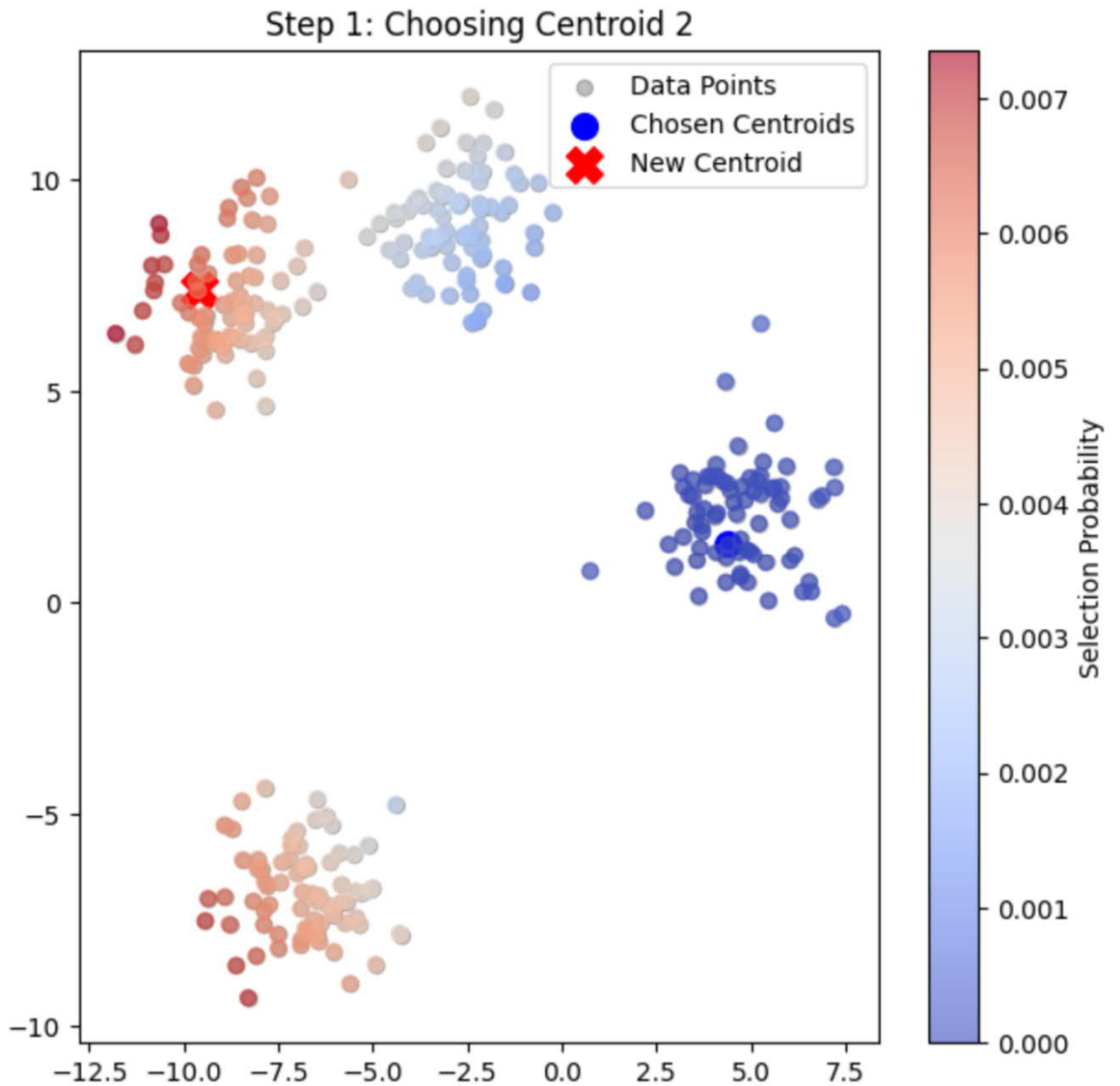
- ▶ Compute probability of choosing x_i as the next centroid:

$$P(x_i) = \frac{D(x_i)^2}{\sum_{x_j \in X} D(x_j)^2}$$

- ▶ Use **weighted random sampling** to pick the next centroid
 - ▶ Points with higher distance $D(x_i)$ have higher chance to be selected.

Example

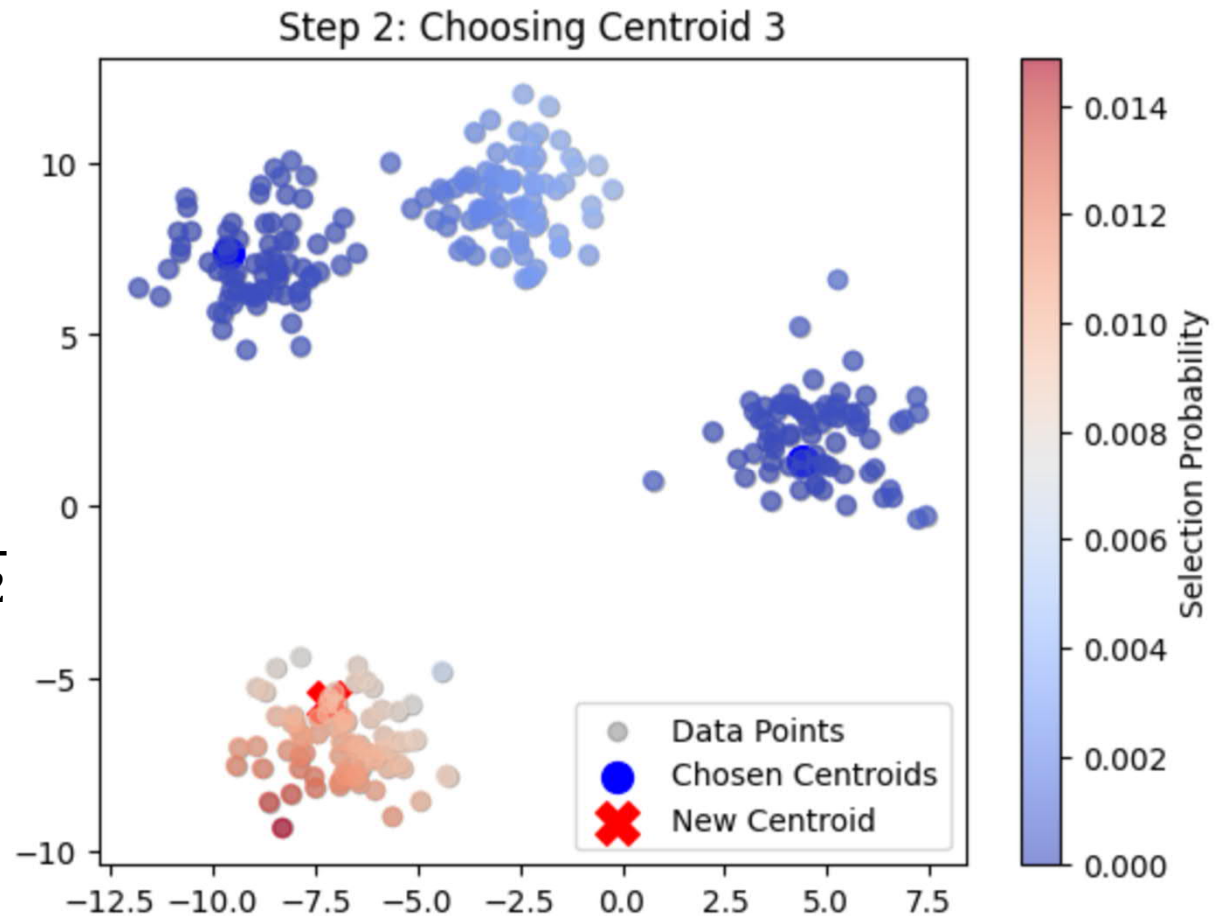
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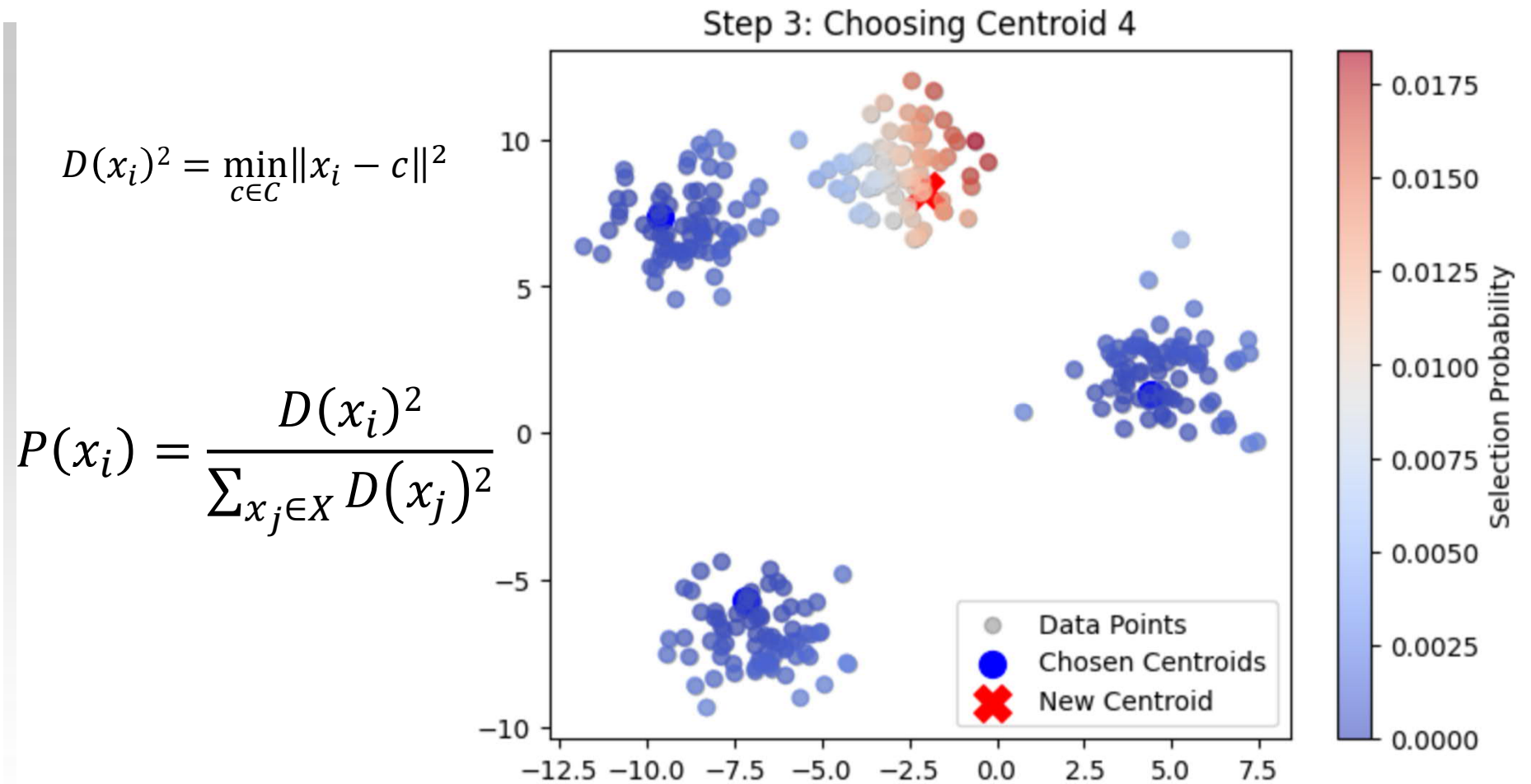
Example

$$D(x_i)^2 = \min_{c \in C} \|x_i - c\|^2$$

$$P(x_i) = \frac{D(x_i)^2}{\sum_{x_j \in X} D(x_j)^2}$$



Example



k-Means++

Why choose subsequent centroids **probabilistically**, why not choosing the x_i with largest $D(x_i)$?

- **Avoid sensitivity to outliers**
- **Better Representation of Data Structure:**
 - Centroids are more likely to be placed in regions of the data that are dense and representative of the underlying clusters.
 - This leads to **better clustering results**.

k-Means vs. k-Means++

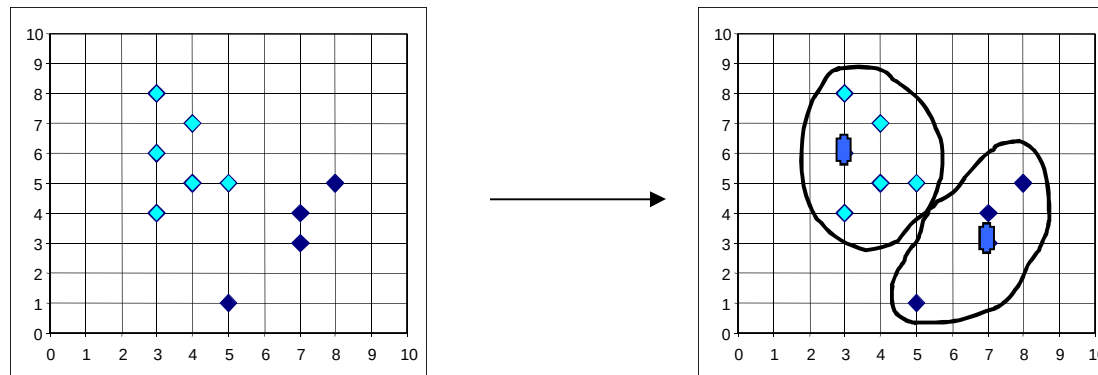
Feature	k-means	k-means++
Centroid Initialization	Random	Distance-based stochastic
Convergence Speed	Can be slow if centroids are bad	Generally faster
Cluster Quality	More likely to get stuck in local minima	More likely to find a better solution
Time complexity	$O(tkn)$	Initialization: $O(kn)$ Subsequent steps: $O(tkn)$ Overall: $O(tkn)$ where t is typically smaller

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Another Problem of k-Means

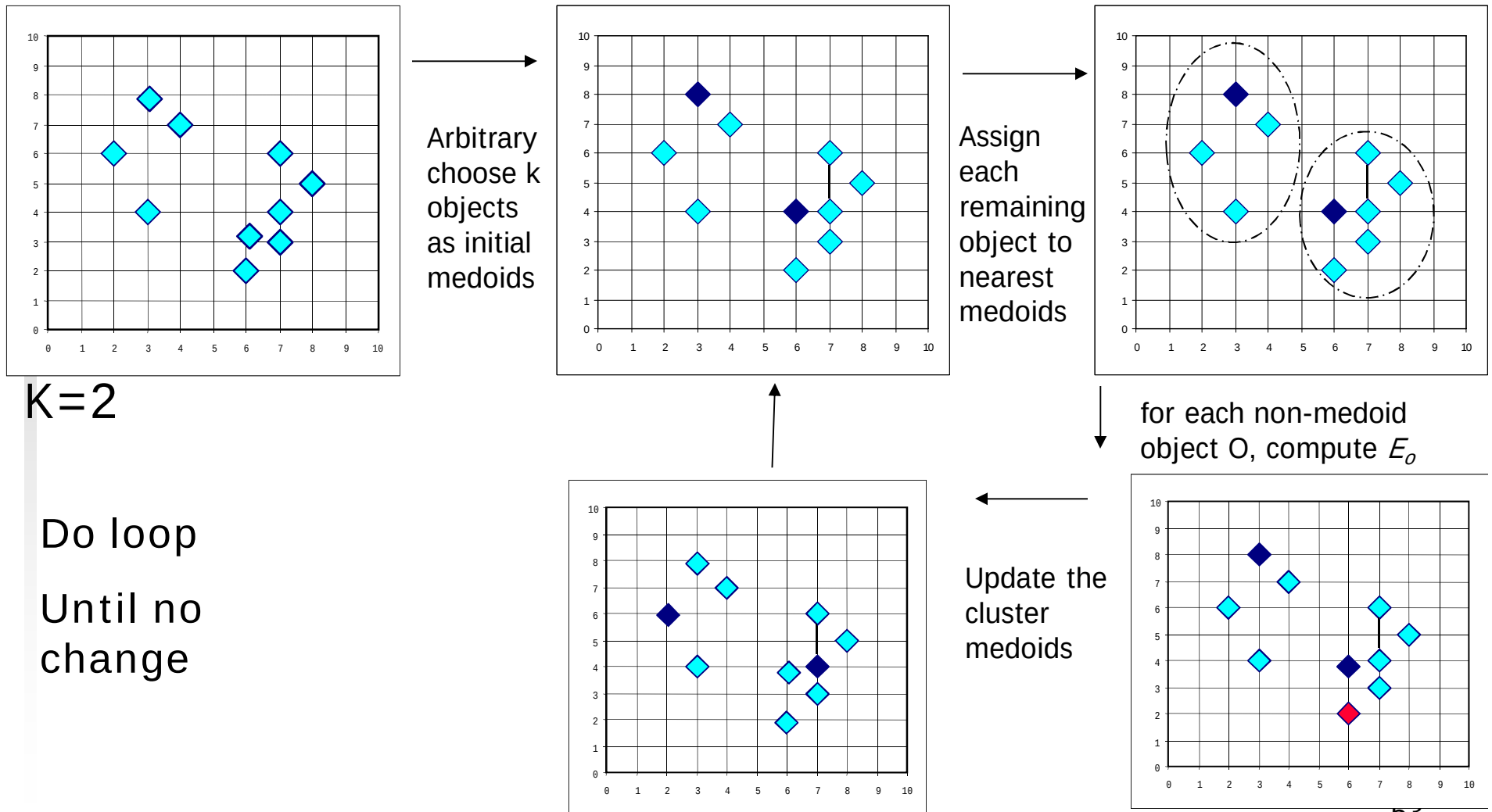
- ▶ The k-means algorithm is sensitive to outliers !
 - ▶ Since an object with an extremely large value may substantially distort the distribution of the data.
- ▶ K-Medoids: Instead of taking the **mean** value of the objects in a cluster as a reference point, **medoids** (also called **clustroids**) can be used, which is the **most centrally located** object in a cluster.



PAM: A K-medoids Method

- ▶ PAM: Partitioning Around Medoids (1987)
- ▶ Arbitrarily choose k objects as the initial medoids
- ▶ Until no change, do
 - ▶ (Re)assign each object to the cluster whose medoid is the nearest
 - ▶ For each pair of a non-medoid object o' and a medoid o , compute the total cost, S , of swapping medoid o with o'
 - ▶ If the lowest cost $S_{lowest} < 0$ then swap o with o_{lowest}' to form the new set of k medoids

Typical k-medoids algorithm (PAM)



How to Choose the new Medoid for a Cluster

- ▶ Use the sum of distances criterion

$$E_o = \sum_{p \in C_i} d(p, o)$$

- o is a candidate medoid for the i th cluster C_i
 - $d(p, o)$ is the distance between o and point p in C_i .
- ▶ Choose point o in C_i that minimizes E_o

Pros and Cons of PAM

- ▶ PAM is more robust than k-means in the presence of noise and outliers
 - ▶ Medoids are less influenced by outliers
- ▶ PAM is efficient for small data sets but does not scale well for large data sets
 - ▶ $O(k(n-k)^2)$ for each iteration
- ▶ Sampling based method: CLARA
 - ▶ Randomly select a subset of data
 - ▶ Using k-medoids to cluster the subset
 - ▶ Remaining examples are assigned to the closest cluster

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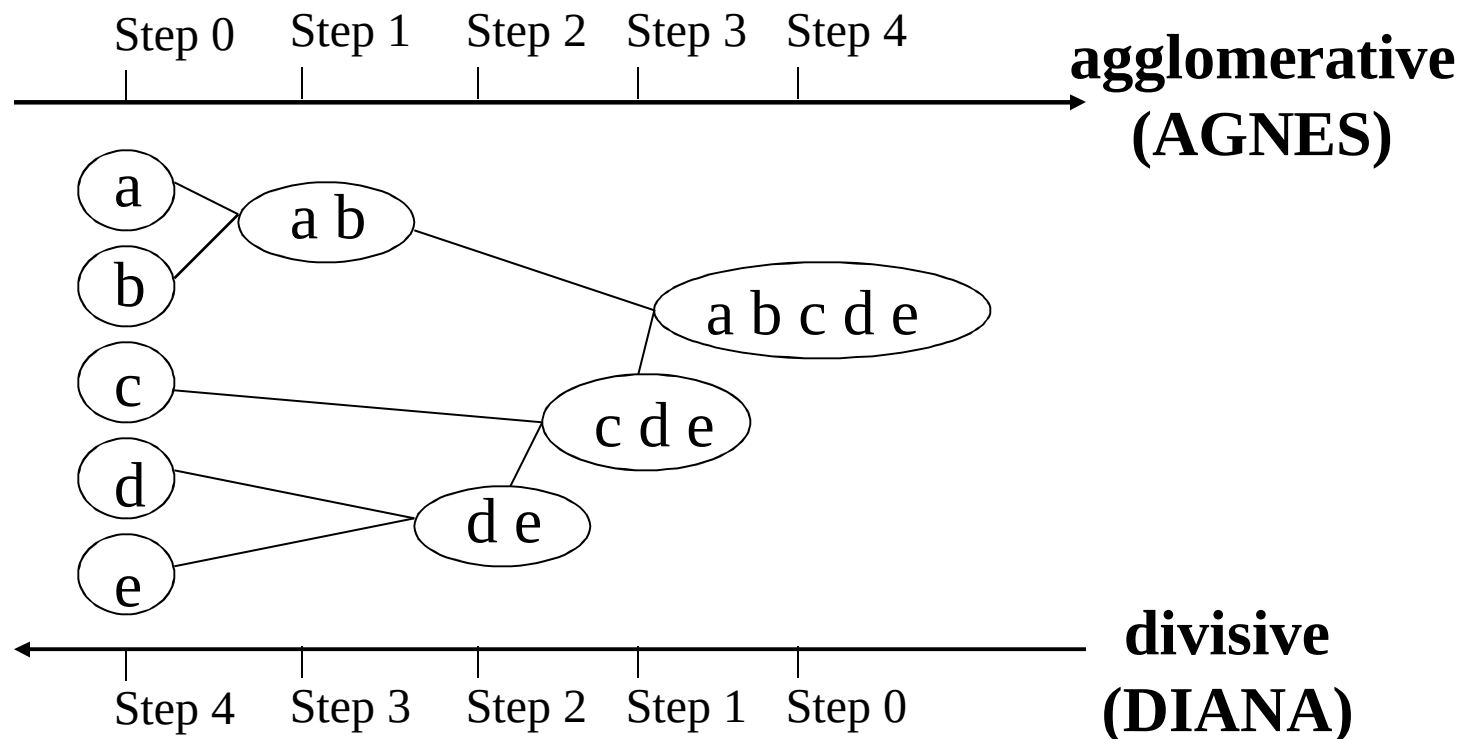
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Hierarchical Clustering

- ▶ Iteratively merge or split clusters to form a tree of clusters
- ▶ Two types
 - ▶ Agglomerative (bottom-up): merge clusters iteratively
 - ▶ Start by placing each object in its own cluster
 - ▶ Merge these small clusters into larger and larger clusters
 - ▶ until all objects are in a single cluster
 - ▶ Most hierarchical methods belong to this category. They differ only in their definition of between-cluster similarity
 - ▶ Divisive (top-down): split a cluster iteratively
 - ▶ Start with all objects in one cluster and subdivide them into smaller pieces

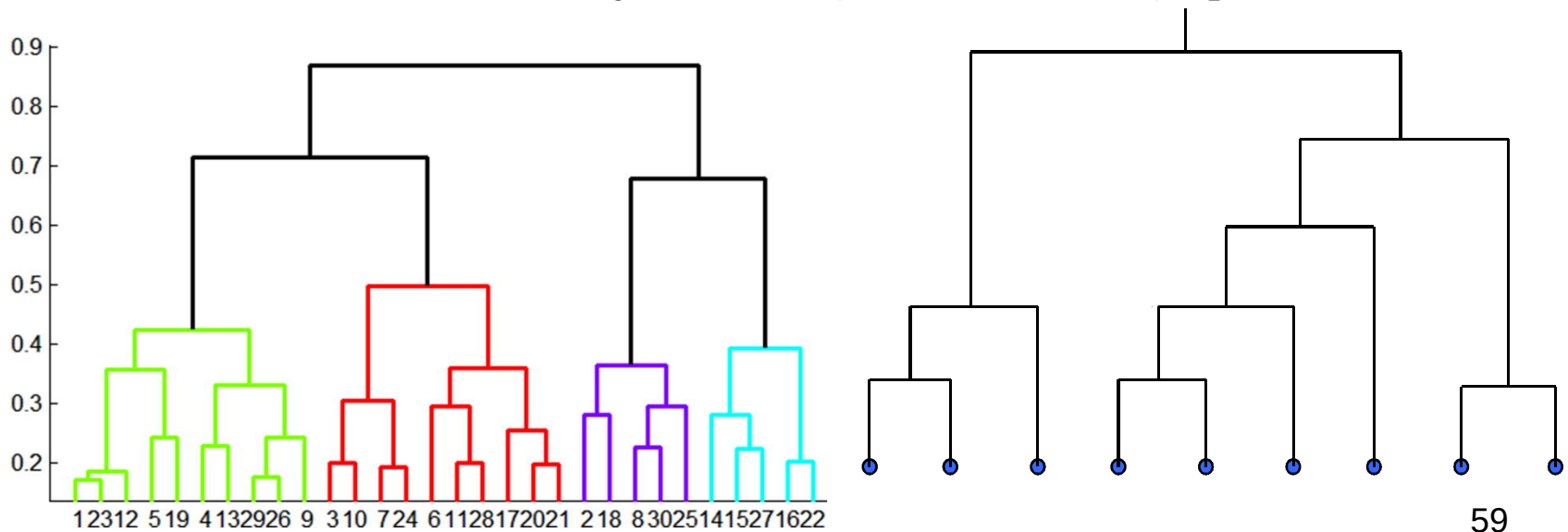
Hierarchical Clustering

- This method **does not require the number of clusters k as an input**, but needs a termination condition

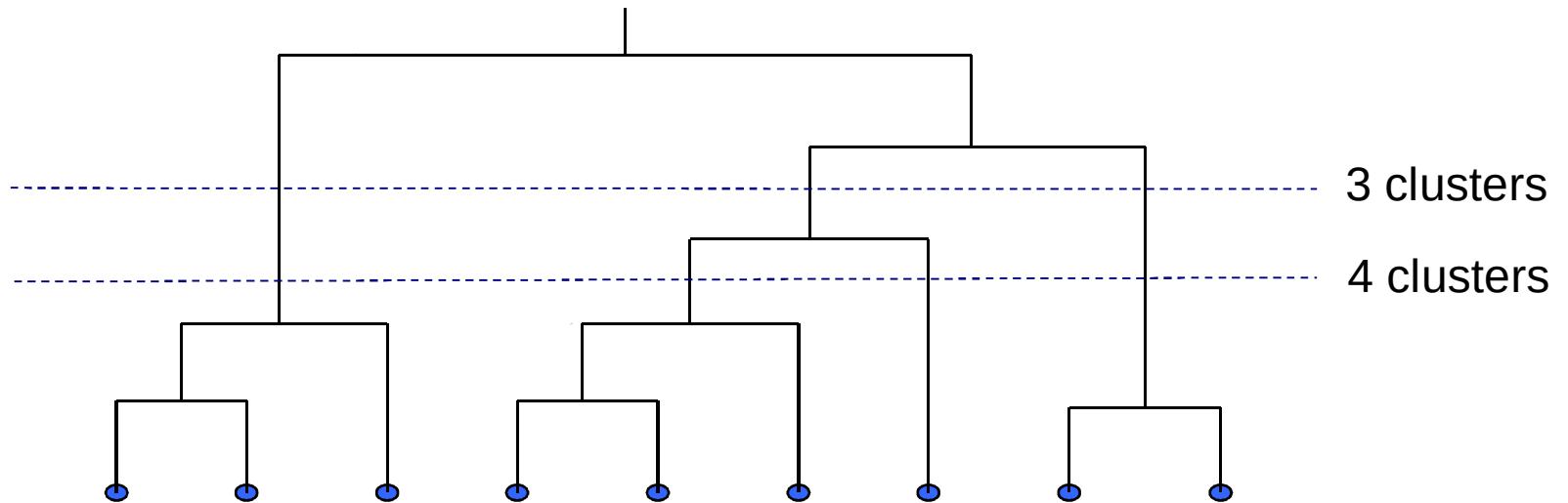


Dendrogram

- ▶ A tree-like diagram that
 - ▶ Visually represents the hierarchical structure of clusters,
 - ▶ Shows how data points or clusters are merged or split at each step of the clustering process.
- ▶ The height of the branches indicates the distance between the merged or split clusters.
 - ▶ Distance should monotonically increase (or non-decrease) upwards



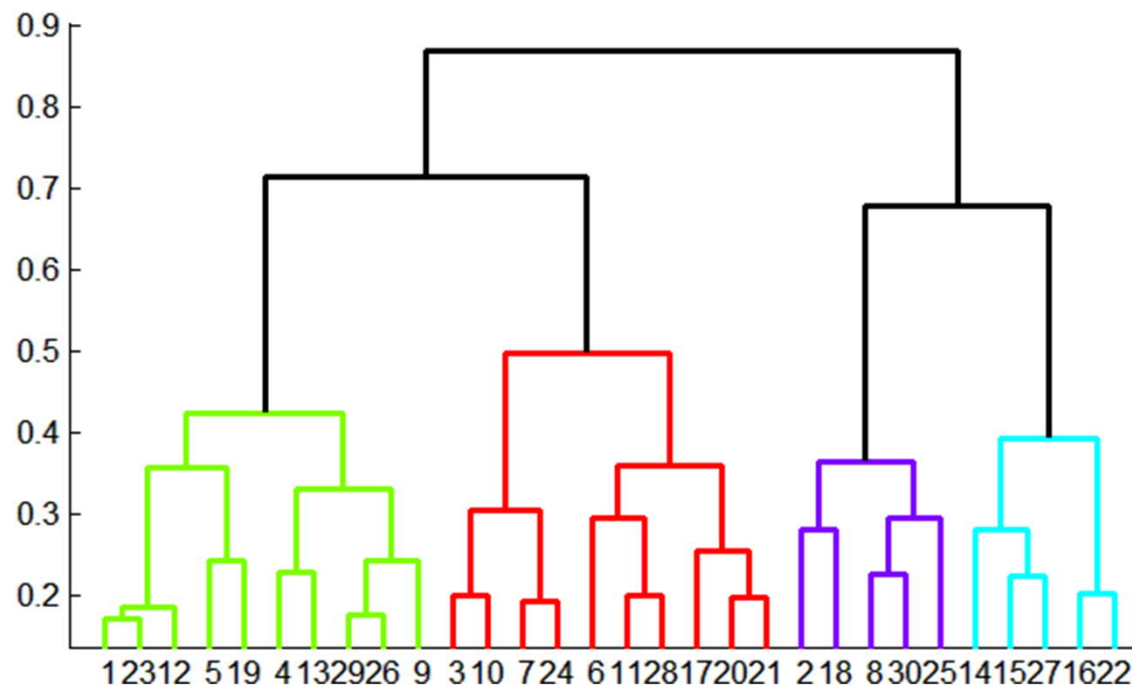
Hierarchical Clustering



- ▶ No need to generate the whole dendrogram. **Merging or splitting clusters can stop:**
 - ▶ if only k clusters are needed, or
 - ▶ Another threshold is satisfied, such as:
 - ▶ The distance between two closest clusters exceeds a distance threshold

Hierarchical Clustering

- ▶ **A Key Question:**
 - ▶ **How do we measure the distance between clusters?**



Inter-cluster Distances in Hierarchical Clustering

- ▶ Several ways to define the distance between two clusters:

- ▶ **Single linkage** method (nearest neighbor):

$$d(C_i, C_j) = \min_{x \in C_i, y \in C_j} \{d(x, y)\}$$

C_i and C_j are
two clusters

- ▶ **Complete linkage** method (furthest neighbor):

$$d(C_i, C_j) = \max_{x \in C_i, y \in C_j} \{d(x, y)\}$$

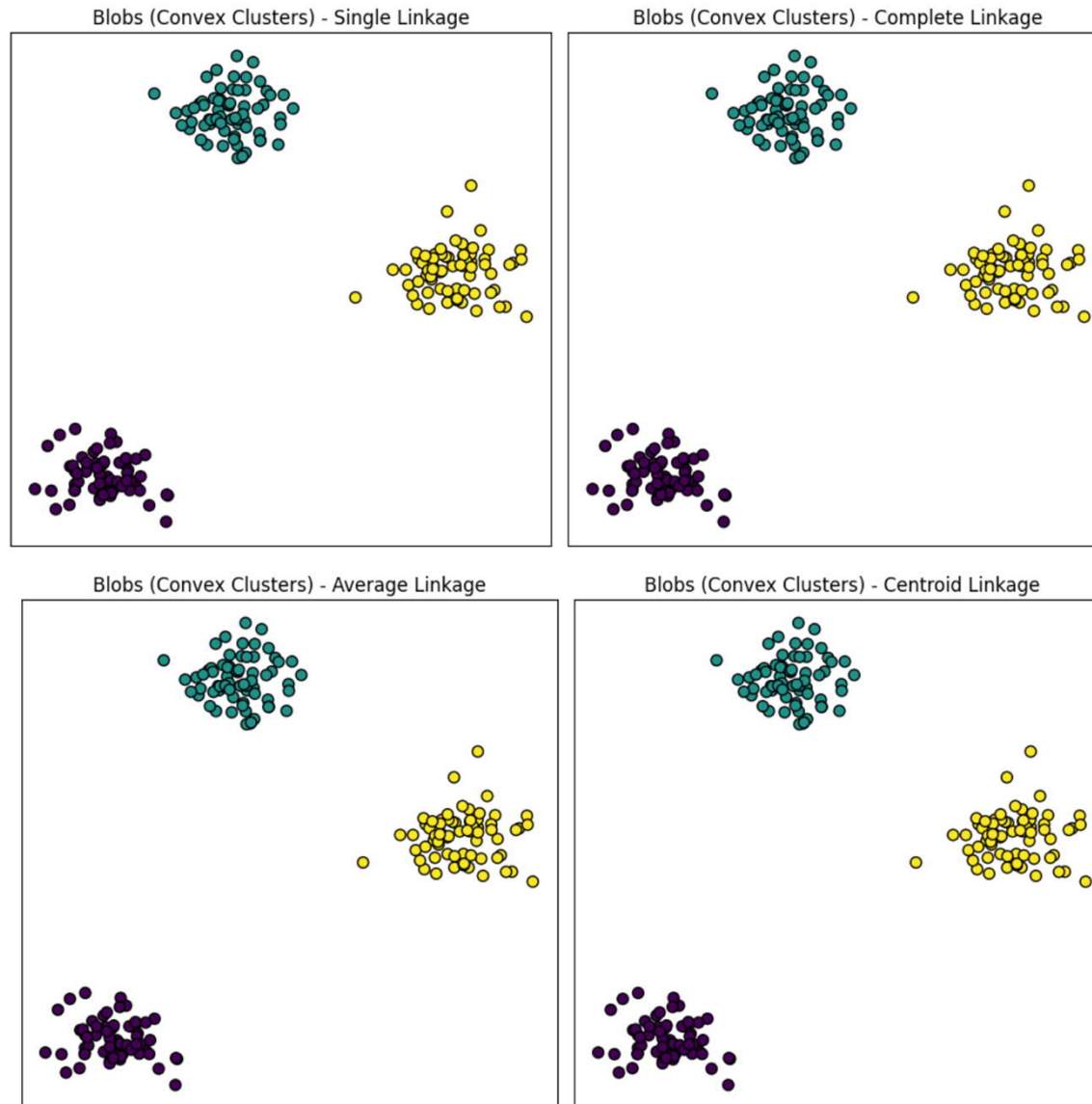
- ▶ **Average linkage** method (unweighted pair-group average):

$$d(C_i, C_j) = \text{avg}_{x \in C_i, y \in C_j} \{d(x, y)\}$$

- ▶ **Centroid linkage** method:

- ▶ Distance between the centroids (e.g., means) of two clusters.

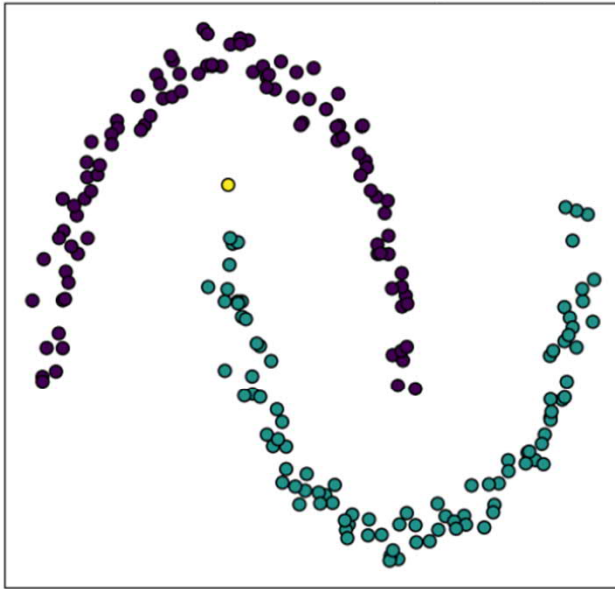
Comparison of Linkage Methods



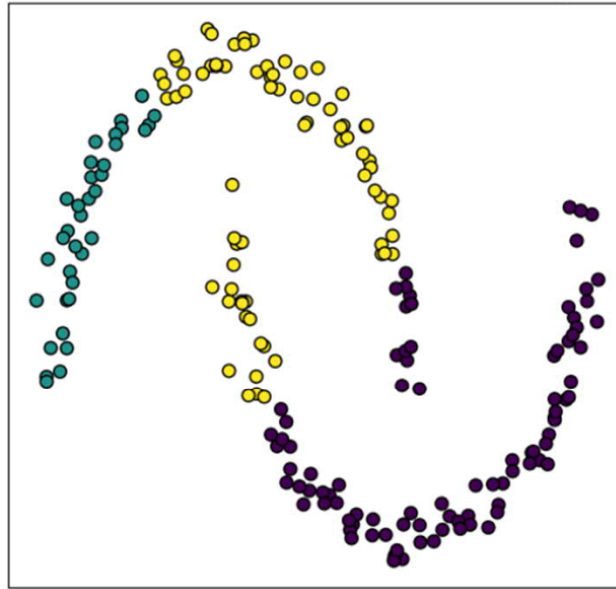
Well-separated, convex clusters

Comparison of Linkage Methods

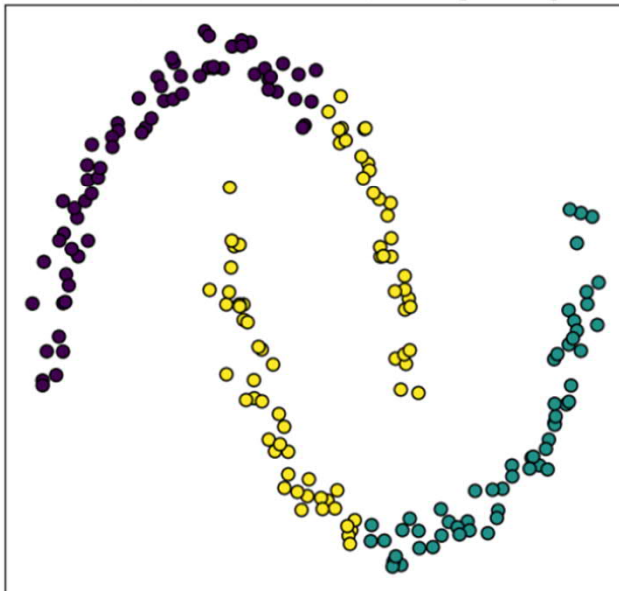
Moons (Non-Convex Clusters) - Single Linkage



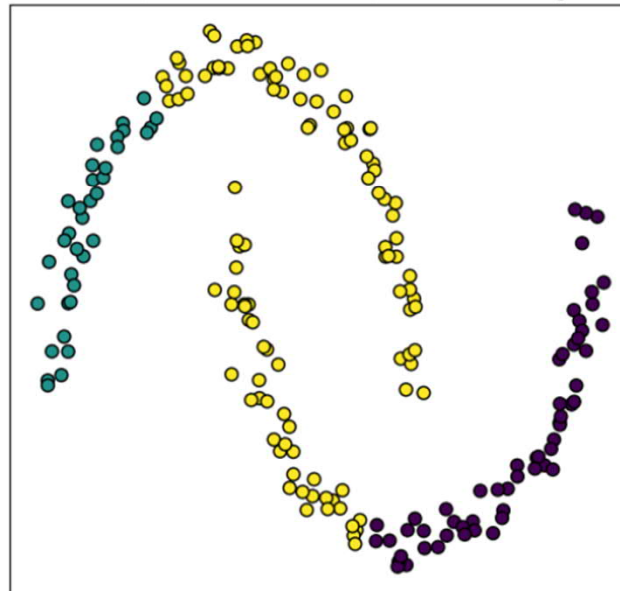
Moons (Non-Convex Clusters) - Complete Linkage



Moons (Non-Convex Clusters) - Average Linkage

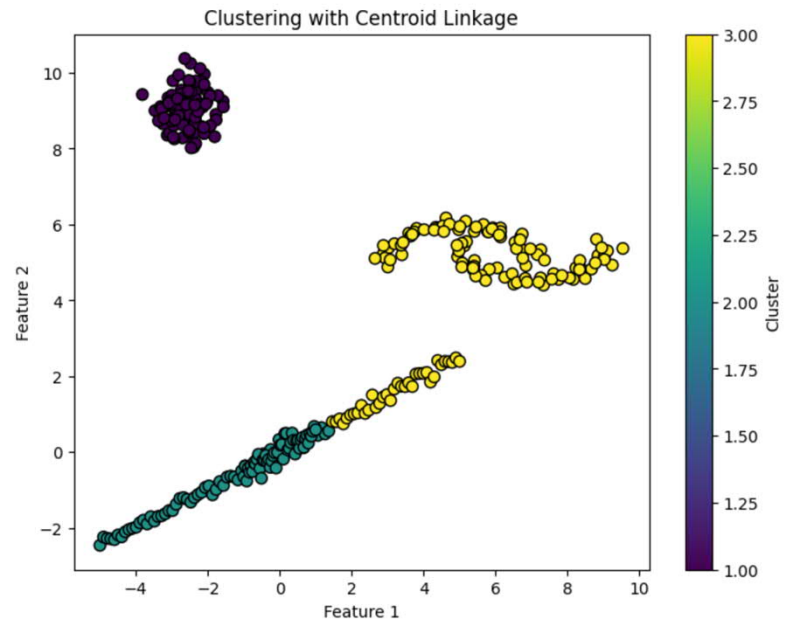
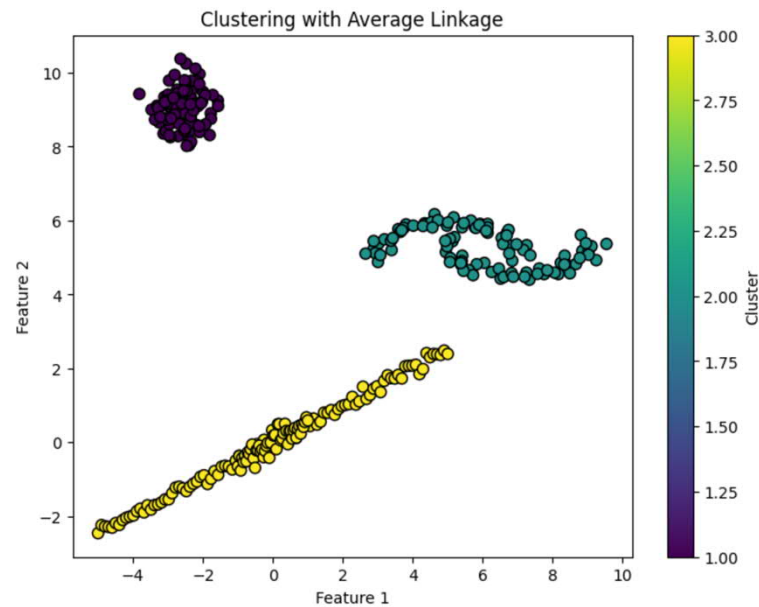
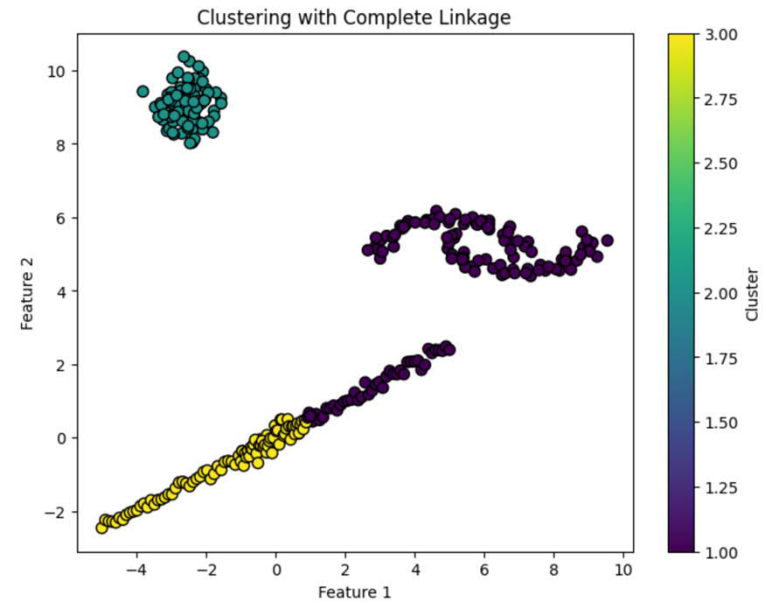
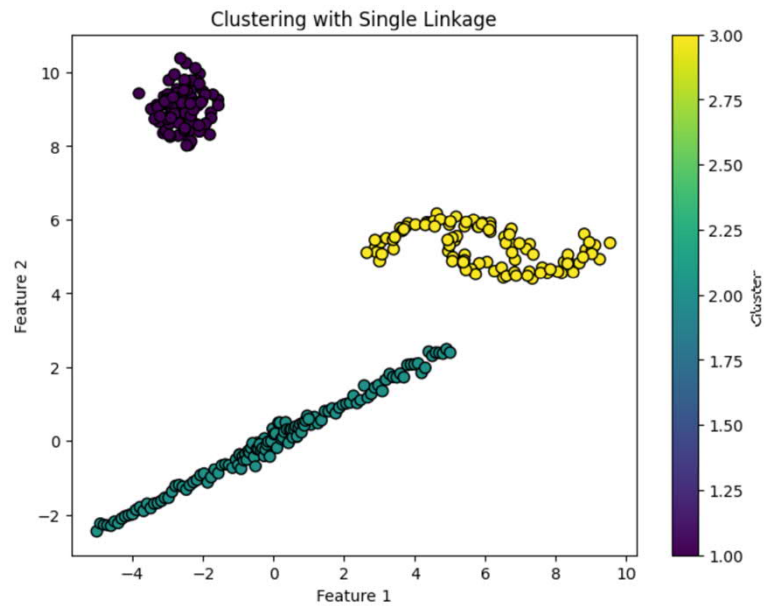


Moons (Non-Convex Clusters) - Centroid Linkage

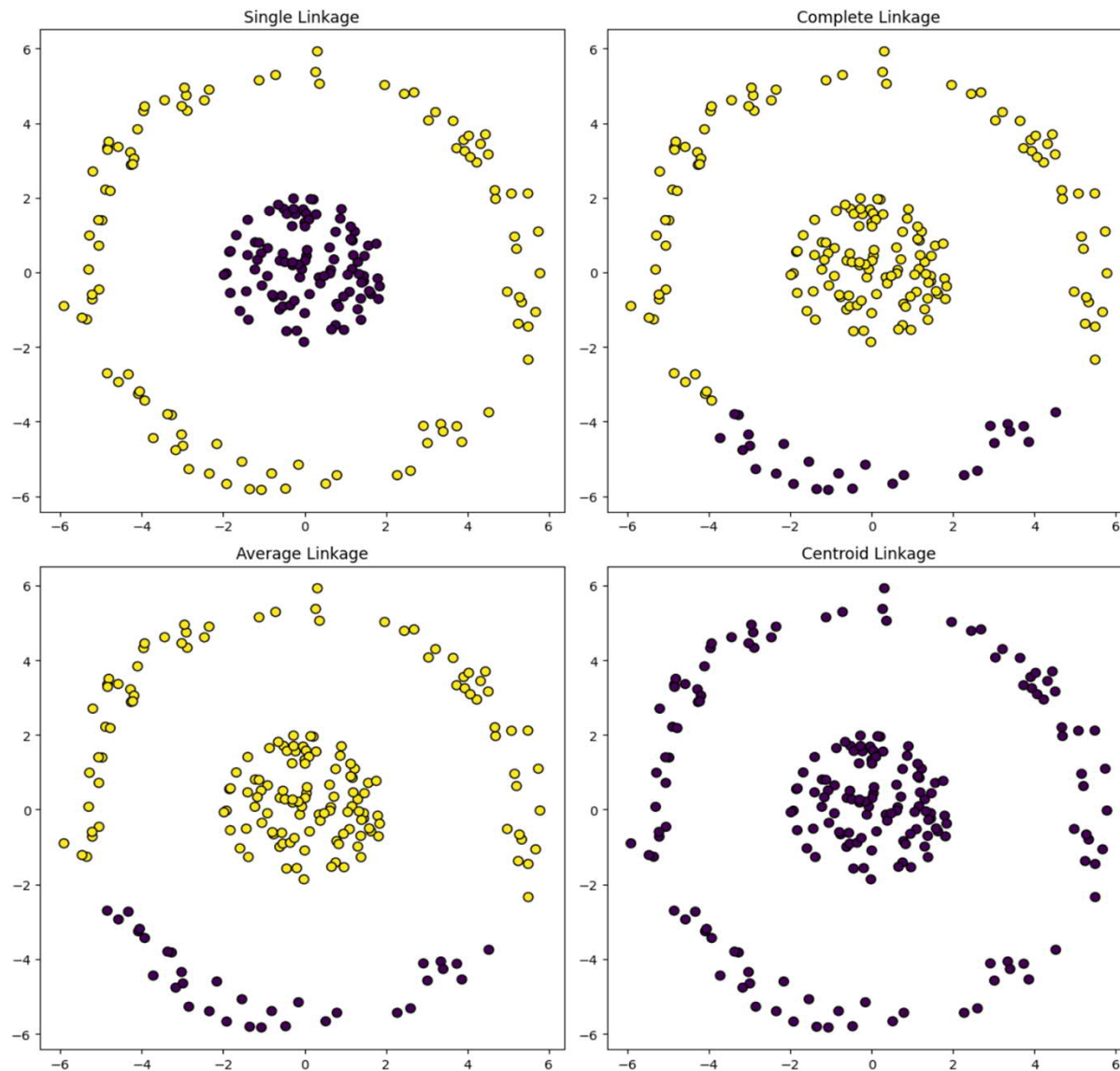


Elongated,
chain-like
clusters

Comparison of Linkage Methods



Comparison of Linkage Methods



Nested
clusters

Comparison of Linkage Methods

- ▶ Single linkage
 - ▶ Good at detecting elongated, chain-like clusters.
 - ▶ **Advantages:**
 - ▶ Can detect non-spherical and non-convex shapes.
 - ▶ **Disadvantages:**
 - ▶ Can suffer from the chaining effect where distant points get merges due to intermediate, noisy points
 - ▶ Not robust to noise between clusters
 - ▶ This type of noise is hard to detect during preprocessing.

Comparison of Linkage Methods

- ▶ Complete linkage
 - ▶ Good at detecting compact, spherical clusters by minimizing the diameter of cluster
 - ▶ Diameter: the distance between the two farthest points in a cluster
 - ▶ **Advantages:**
 - ▶ Robust to between-cluster noise and outliers compared to single linkage.
 - ▶ Tends to produce more balanced clusters.
 - ▶ **Disadvantages:**
 - ▶ Sensitive to outliers within clusters that are far away; may fail to merge clusters that should be merged.
 - ▶ May break large clusters into smaller ones.
 - ▶ Struggles with detecting clusters of varying sizes or non-spherical shapes.

Comparison of Linkage Methods

▶ Average linkage

- ▶ A compromise between single and complete linkage, suitable for detecting clusters of various shapes and sizes.
 - ▶ But not as flexible as single linkage in terms of cluster shapes
- ▶ **Advantages:**
 - ▶ Less sensitive to noise and outliers (due to averaging out the effect of outliers).
 - ▶ Can handle clusters of different sizes and shapes better than complete and centroid linkage methods.
- ▶ **Disadvantages:**
 - ▶ Computationally more expensive than single and complete linkage.
 - ▶ Need to recompute the average distances between the merged cluster and each of others after merging or splitting.
 - ▶ May still struggle with very irregular cluster shapes.

Comparison of Linkage Methods

▶ Centroid linkage

- ▶ Detecting spherical clusters, similar to k-means

▶ **Advantages:**

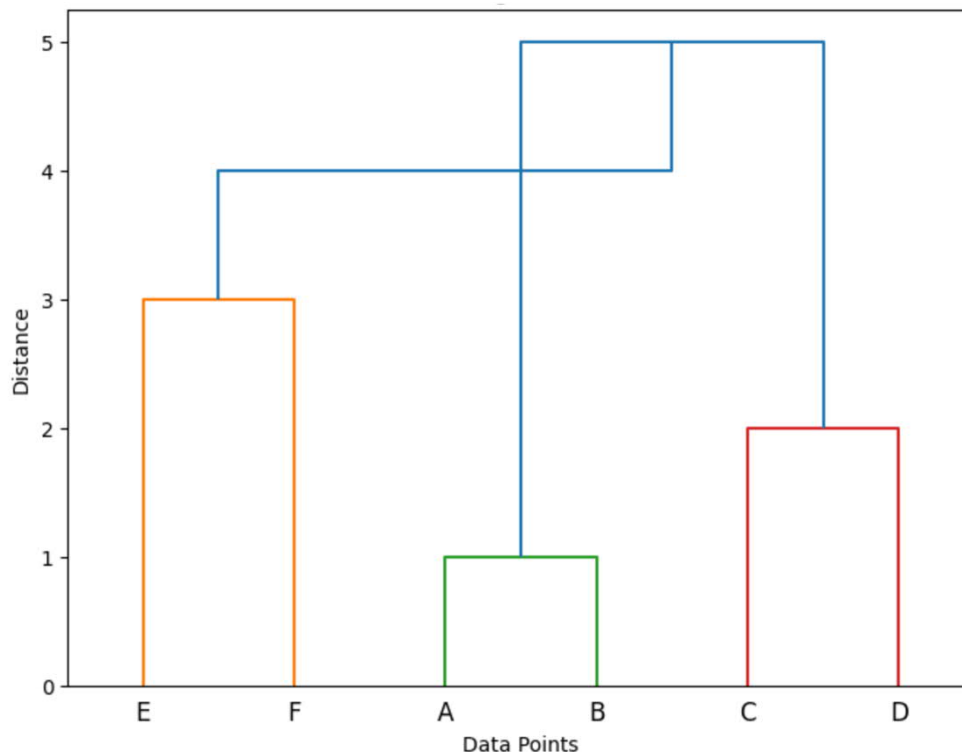
- ▶ Intuitive and easy to understand.
- ▶ Works well with clusters that are roughly spherical and of similar size.

▶ **Disadvantages:**

- ▶ Struggles with clusters that have irregular or non-convex shapes (e.g., crescent moons, rings, or chains).
 - ▶ Centroids can be out of non-convex shaped clusters (not a good cluster representation)
- ▶ Sensitive to outliers
 - ▶ Outliers or sparse regions in a cluster can pull the centroid away from the dense region, distorting the distance calculations
- ▶ Can produce "inversions" in the dendrogram, where the distance between merged clusters decreases instead of increasing.
 - ▶ This happens because the centroid of the merged cluster might be closer to another cluster than the original clusters were.

Dendrogram with Inversions

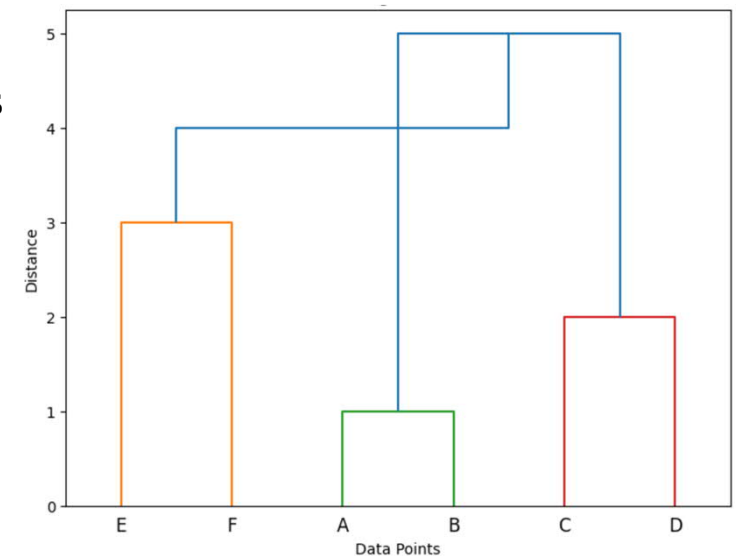
- ▶ Distance may decrease instead of increasing upwards.
- ▶ Example:
 - ▶ Suppose we have 6 data points: A, B, C, D, E, F. We perform hierarchical clustering using centroid linkage



- The second last merge of cluster AUB and cluster CUD is at distance $d=5$
- The last merge of cluster (EUF) and cluster (AUBUCUD) is at distance $d=4$
- Distance decreases

Dendrogram with Inversions

- ▶ Cause of inversions:
 - ▶ The centroid of the merged cluster might be closer to another cluster than the original clusters were.
- ▶ Problems it may cause
 - ▶ Difficulty interpreting the dendrogram
 - ▶ Inversions make the dendrogram non-monotonic, as height increases.
 - ▶ Lead to confusion when trying to determine the relationships between clusters.
 - ▶ Challenges in obtaining the final clusters:
 - ▶ When cutting the dendrogram to obtain a specific number of clusters, inversions can make it unclear where to cut, as the distances may not reflect the true relationships between clusters.
- ▶ This is why centroid linkage is **not commonly used** in hierarchical clustering



Summary of When to Use What

- ▶ Single Linkage:
 - ▶ Use when you expect elongated clusters and can handle noise.
- ▶ Complete Linkage:
 - ▶ Use when you expect compact, well-separated clusters.
- ▶ Average Linkage:
 - ▶ Use as a general-purpose method for clusters of varying shapes and sizes.
- ▶ Centroid Linkage:
 - ▶ Not commonly used
 - ▶ Can be used when clusters are expected to be spherical and of similar size, but be cautious of inversions.

Strengths of Hierarchical Clustering

- ▶ Conceptually simple
- ▶ No need to specify the number of clusters
- ▶ Captures hierarchical relationships
- ▶ Flexibility in Cluster Shapes:
 - ▶ Depending on the linkage method, hierarchical clustering can handle clusters of varying shapes and sizes, though it is not as robust as density-based methods like DBSCAN.
- ▶ Deterministic
 - ▶ Produce the same result every time for the same dataset.
- ▶ Visualization:
 - ▶ The dendrogram provides an intuitive visualization of the clustering process, making it easier to interpret the results.

Limitations of Hierarchical Clustering

- ▶ Do not scale well:
 - ▶ Time complexity is $O(n^3)$ for naïve implementation and $O(n^2 \log n)$ for optimized implementations, where n is the number of total data points
- ▶ Sensitivity to Noise and Outliers:
 - ▶ Some linkage methods (e.g., single linkage) are sensitive to noise and outliers, which can lead to poor clustering results.
- ▶ Irreversible Merges:
 - ▶ Once two clusters are merged/split, the decision cannot be reversed. This can lead to suboptimal clustering if early merges are incorrect.
- ▶ Choice of Linkage Method:
 - ▶ The results depend heavily on the choice of linkage method, and selecting the wrong method can lead to poor clustering.